

MATH4045GU

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ABSTRACT. Lecture notes for MATH4045GU.

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¹Based on lectures by Yoonjoo Kim [1].

²Based on lectures by Robert L. Bryant [2].

Part 1. Algebraic Curves³

1. RIEMANN SURFACES AND PLANE CURVES

A domain is connected and open, and $V(F)$ denotes the zero locus of F .

1.1. Complex Analysis.

Definition 1.1.1 (Holomorphicity). Let $V \subset \mathbb{C}$ be open, $z = x + iy$, and $f = u + iv: V \rightarrow \mathbb{C}$. Then

$$f'(z_0) := \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h}, \quad f \in \mathcal{O}(V) \iff f'(z) \text{ exists for every } z \in V.$$

The Wirtinger operators are

$$\frac{\partial}{\partial z} := \frac{1}{2} \left(\frac{\partial}{\partial x} - i \cdot \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} := \frac{1}{2} \left(\frac{\partial}{\partial x} + i \cdot \frac{\partial}{\partial y} \right).$$

Proposition 1.1.2 (Cauchy–Riemann). For $f = u + iv \in C^1(V)$,

$$f \in \mathcal{O}(V) \iff \frac{\partial}{\partial x} u = \frac{\partial}{\partial y} v \wedge \frac{\partial}{\partial y} u = -\frac{\partial}{\partial x} v \iff \frac{\partial}{\partial \bar{z}} f = 0, \quad f' = \frac{\partial}{\partial z} f.$$

Put

$$J := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad J_{\mathbb{R}}(f) := \begin{bmatrix} \frac{\partial}{\partial x} u & \frac{\partial}{\partial y} u \\ \frac{\partial}{\partial x} v & \frac{\partial}{\partial y} v \end{bmatrix}.$$

Equivalently,

$$\begin{array}{ccc} T_z \mathbb{C} & \xrightarrow{df_z} & T_{f(z)} \mathbb{C} \\ J \downarrow & & \downarrow J \\ T_z \mathbb{C} & \xrightarrow{df_z} & T_{f(z)} \mathbb{C}. \end{array}$$

$J_{\mathbb{R}}(f) \cdot J = J \cdot J_{\mathbb{R}}(f),$

Proof. Compare the two coordinate-axis difference quotients and the real differential.

$$\lim_{t \rightarrow 0} \frac{f(z+t) - f(z)}{t} = \frac{\partial}{\partial x} u + i \cdot \frac{\partial}{\partial x} v, \quad \lim_{t \rightarrow 0} \frac{f(z+it) - f(z)}{it} = \frac{\partial}{\partial y} v - i \cdot \frac{\partial}{\partial y} u.$$

$$\frac{\partial}{\partial x} u + i \cdot \frac{\partial}{\partial x} v = \frac{\partial}{\partial y} v - i \cdot \frac{\partial}{\partial y} u \iff \frac{\partial}{\partial x} u = \frac{\partial}{\partial y} v \wedge \frac{\partial}{\partial y} u = -\frac{\partial}{\partial x} v.$$

$$\begin{bmatrix} \frac{\partial}{\partial x} u & \frac{\partial}{\partial y} u \\ \frac{\partial}{\partial x} v & \frac{\partial}{\partial y} v \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial}{\partial x} u & \frac{\partial}{\partial y} u \\ \frac{\partial}{\partial x} v & \frac{\partial}{\partial y} v \end{bmatrix} \iff \frac{\partial}{\partial \bar{z}} f = 0.$$

$$f(z+h) - f(z) = \left(\frac{\partial}{\partial z} f \right)(z) \cdot h + \left(\frac{\partial}{\partial \bar{z}} f \right)(z) \cdot \bar{h} + o(|h|).$$

$$\left(\frac{\partial}{\partial \bar{z}} f \right)(z) = 0 \implies \frac{f(z+h) - f(z)}{h} = \left(\frac{\partial}{\partial z} f \right)(z) + o(1).$$

Thus all four formulations are equivalent. □

³Based on lectures by Yoonjoo Kim [1].

Theorem 1.1.3 (Cauchy formula). *Let $\gamma \subset V$ be a positively oriented piecewise- C^1 Jordan curve with $\overline{\Omega_\gamma} \subset N_\gamma \Subset V$. For $f \in \mathcal{O}(V)$, $z_0 \in \Omega_\gamma$, and $n \geq 0$,*

$$(1.1.1) \quad f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z-z_0)^{n+1}} dz, \quad f^{(0)} := f.$$

Writing the integral in (1.1.1) as $\mathcal{C}_{n,z_0}$ gives

$$\begin{array}{ccc} \mathcal{O}(V) & \xrightarrow{\text{res}} & \mathcal{O}(N_\gamma) \\ & \searrow f \mapsto f^{(n)}(z_0) & \downarrow \mathcal{C}_{n,z_0} \\ & & \mathbb{C}. \end{array}$$

Corollary 1.1.4 (Taylor and Laurent expansions). *For $0 < \rho < R$,*

$$f \in \mathcal{O}(B_R(z_0)) \implies f(z) = \sum_{n \geq 0} a_n (z - z_0)^n, \quad a_n = \frac{f^{(n)}(z_0)}{n!},$$

$$f \in \mathcal{O}(B_R(z_0) \setminus \overline{B_\rho(z_0)}) \implies f(z) = \sum_{n \in \mathbb{Z}} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_{|w-z_0|=r} \frac{f(w)}{(w-z_0)^{n+1}} dw$$

for every $\rho < r < R$.

Theorem 1.1.5 (Isolated singularities). *Let $V \subset \mathbb{C}$ be open, $z_0 \in V$, and $f \in \mathcal{O}(V \setminus \{z_0\})$. For*

$$P_f := \sum_{n < 0} a_n (z - z_0)^n,$$

exactly one branch occurs:

$$\begin{array}{lcl} f & \nearrow & P_f = 0 \xrightarrow{\text{removable}} f \text{ extends across } z_0 \\ & \longrightarrow & 0 < \#\{n < 0 : a_n \neq 0\} < \infty \xrightarrow{\text{pole of order } m} f = (z - z_0)^{-m} \cdot h, \quad h(z_0) \neq 0 \\ & \searrow & \#\{n < 0 : a_n \neq 0\} = \infty \xrightarrow{\text{essential}} z_0 \text{ is essential.} \end{array}$$

Here $m \in \mathbb{Z}_{>0}$, h is holomorphic near z_0 , and the removable branch is equivalent to $f = O(1)$ as $z \rightarrow z_0$.

Theorem 1.1.6 (Identity theorem). *Let $V \subset \mathbb{C}$ be a domain and $f, g \in \mathcal{O}(V)$. If $S \subset V$ has an accumulation point in V , then*

$$f|_S = g|_S \implies f = g \text{ on } V, \quad \begin{array}{ccc} S & \hookrightarrow & V \\ \downarrow (f,g)|_S & \searrow (f,f) & \downarrow (f,g) \\ \Delta_{\mathbb{C}} & \hookrightarrow & \mathbb{C}^2. \end{array}$$

Here $\Delta_{\mathbb{C}} := \{(a, a) : a \in \mathbb{C}\}$.

Theorem 1.1.7 (Open mapping and maximum modulus). *For $f \in \mathcal{O}(V)$ and every nonempty open $U \subset V$,*
 f nonconstant $\implies f(U)$ open, $|f|$ has an interior local maximum $\implies f$ is constant.

$$\begin{array}{ccc}
 U & \hookrightarrow & V \\
 f|_U \downarrow & & \downarrow f \\
 f(U) & \hookrightarrow & \mathbb{C}
 \end{array}
 \qquad
 \begin{array}{ccc}
 V & \xrightarrow{f} & \mathbb{C} \\
 & \searrow |f| & \downarrow |\cdot| \\
 & & \mathbb{R}_{\geq 0}.
 \end{array}$$

Theorem 1.1.8 (Holomorphic inverse theorem). *If $f \in \mathcal{O}(V)$ and $f'(z_0) \neq 0$, then some neighborhoods $z_0 \in U \subset V$ and $f(z_0) \in W \subset \mathbb{C}$ fit into*

$$\begin{array}{ccccc}
 (U, z_0) & \xrightarrow{f} & (W, f(z_0)) & \xrightarrow{f^{-1}} & (U, z_0), \\
 & \xleftarrow{f^{-1}} & & & \\
 & \searrow \text{id}_U & & & \\
 & & & &
 \end{array}
 \qquad
 \begin{array}{ccc}
 T_z \mathbb{C} & \xrightarrow{df_z} & T_{f(z)} \mathbb{C} \\
 & \searrow \text{id} & \downarrow d(f^{-1})_{f(z)} \\
 & & T_z \mathbb{C}.
 \end{array}$$

$$(f^{-1})'(w) = \frac{1}{f'(f^{-1}(w))}.$$

Corollary 1.1.9 (Liouville). *For $f \in \mathcal{O}(\mathbb{C})$, $C > 0$, and $N \in \mathbb{Z}_{\geq 0}$,*

$$|f(z)| \leq C \cdot (1 + |z|)^N \implies f \in \mathbb{C}[z] \wedge \deg(f) \leq N, \quad N = 0 \implies f \text{ is constant}.$$

1.2. Riemann Surfaces.

Definition 1.2.1 (Riemann surface). A Riemann surface is a connected second-countable Hausdorff space X with a maximal atlas

$$\phi_\alpha: U_\alpha \xrightarrow{\sim} V_\alpha \subset \mathbb{C}$$

such that

$$\phi_{\beta\alpha} := \phi_\beta \circ \phi_\alpha^{-1} \in \mathcal{O}(\phi_\alpha(U_\alpha \cap U_\beta)), \quad \frac{\partial}{\partial \bar{z}} \phi_{\beta\alpha} \neq 0.$$

For $U_{\alpha\beta\gamma} := U_\alpha \cap U_\beta \cap U_\gamma$,

$$\begin{array}{ccc}
 \phi_\alpha(U_{\alpha\beta\gamma}) & \xrightarrow{\phi_{\beta\alpha}} & \phi_\beta(U_{\alpha\beta\gamma}) \\
 & \searrow \phi_{\gamma\alpha} & \downarrow \phi_{\gamma\beta} \\
 & & \phi_\gamma(U_{\alpha\beta\gamma}),
 \end{array}
 \qquad
 \phi_{\gamma\alpha} = \phi_{\gamma\beta} \circ \phi_{\beta\alpha}.$$

Proposition 1.2.2 (Canonical orientation). *Every Riemann surface is canonically an oriented smooth real surface. On an overlap, writing $\phi_{\beta\alpha} = u + iv$,*

$$J_{\mathbb{R}}(\phi_{\beta\alpha}) = \begin{bmatrix} \frac{\partial}{\partial x} u & \frac{\partial}{\partial y} u \\ \frac{\partial}{\partial x} v & \frac{\partial}{\partial y} v \end{bmatrix} \xrightarrow{1.1.2} \begin{bmatrix} \frac{\partial}{\partial x} u & -\frac{\partial}{\partial x} v \\ \frac{\partial}{\partial x} v & \frac{\partial}{\partial x} u \end{bmatrix},$$

$$\det(J_{\mathbb{R}}(\phi_{\beta\alpha})) = \left(\frac{\partial}{\partial x} u \right)^2 + \left(\frac{\partial}{\partial x} v \right)^2 = \left| \frac{\partial}{\partial \bar{z}} \phi_{\beta\alpha} \right|^2 > 0.$$

Corollary 1.2.3 (Genus). *For compact X , there is a unique $g(X) \in \mathbb{Z}_{\geq 0}$ and an orientation-preserving homeomorphism*

$$X \cong \Sigma_{g(X)}, \quad \chi(X) = 2 - 2g(X).$$

Definition 1.2.4 (Complex projective space). For $n \geq 1$,

$$\mathbb{P}^n := (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^\times, \quad U_i := \{[x_0 : \cdots : x_n] : x_i \neq 0\},$$

$$\phi_i([x_0 : \cdots : x_n]) = \left(\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right).$$

For $q_i(x) := \phi_i([x])$,

$$\begin{array}{ccc} \{x \in \mathbb{C}^{n+1} : x_i \neq 0\} & \xrightarrow{q} & U_i \\ & \searrow q_i & \downarrow \sim \phi_i \\ & & \mathbb{C}^n. \end{array}$$

Example 1.2.5 (Projective line and complex tori). The projective-line charts and the lattice quotient satisfy

$$\begin{array}{ccc} U_0 \cap U_\infty & \xrightarrow[\sim]{\phi_0} & \mathbb{C}^\times \\ \phi_\infty \downarrow \sim & & \downarrow t \mapsto t^{-1} \\ \mathbb{C}^\times & \xlongequal{\quad} & \mathbb{C}^\times \end{array} \quad \begin{array}{ccc} \mathbb{C} & \xrightarrow{T_\lambda(z)=z+\lambda} & \mathbb{C} \\ \pi \downarrow & \searrow \pi & \downarrow \pi \\ \mathbb{C}/L & \xlongequal{\quad} & \mathbb{C}/L. \end{array}$$

Here

$$\begin{aligned} \phi_0([z : w]) &= w/z, & \phi_\infty([z : w]) &= z/w, & \mathbb{P}^1 &\cong S^2, & g(\mathbb{P}^1) &= 0, \\ L &= \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2, & 0 < 2\varepsilon &< \inf_{0 \neq \lambda \in L} |\lambda|, \end{aligned}$$

$$\phi_{[z_0]} : \pi(B_\varepsilon(z_0)) \xrightarrow{\sim} B_\varepsilon(0), \quad \phi_{[z_0]}(\pi(z)) = z - z_0.$$

On every connected overlap,

$$\phi_{[z_1]} \circ \phi_{[z_0]}^{-1}(t) = t + z_0 - z_1 - \lambda, \quad \lambda \in L.$$

Consequently,

$$\mathbb{C}/L \text{ is compact}, \quad \mathbb{C}/L \cong T^2, \quad g(\mathbb{C}/L) = 1.$$

Definition 1.2.6 (Holomorphic functions). For $W \subset X$ open and $f : W \rightarrow \mathbb{C}$,

$$f \in \mathcal{O}_X(W) \iff f \circ \phi^{-1} \in \mathcal{O}(\phi(U)) \quad \text{for every chart } \phi : U \xrightarrow{\sim} \phi(U), \quad U \subset W.$$

On an overlap,

$$\begin{array}{ccc} \phi_\alpha(U_{\alpha\beta}) & \xrightarrow{\phi_{\beta\alpha}} & \phi_\beta(U_{\alpha\beta}) \\ & \searrow f \circ \phi_\alpha^{-1} & \downarrow f \circ \phi_\beta^{-1} \\ & & \mathbb{C}, \end{array} \quad f \circ \phi_\alpha^{-1} = (f \circ \phi_\beta^{-1}) \circ \phi_{\beta\alpha}.$$

Corollary 1.2.7 (Identity theorem). For a domain $W \subset X$ and $f, g \in \mathcal{O}_X(W)$,

$$\{p \in W : f(p) = g(p)\} \text{ has an accumulation point} \xrightarrow{1.1.6} f = g \text{ on } W.$$

Proposition 1.2.8 (Compact maximum). If X is compact, then

$$\mathcal{O}_X(X) = \mathbb{C}.$$

Proof. Choose $p_0 \in X$ maximizing $|f|$ and a connected chart $\phi : U \rightarrow V$ at p_0 .

$$\begin{array}{ccccc}
 U & \xrightarrow{\phi} & V & \xrightarrow{f \circ \phi^{-1}} & \mathbb{C}. \\
 & \searrow & \uparrow & \nearrow & \\
 & & f|_U & &
 \end{array}$$

$$|f(p_0)| \geq |f(p)| \quad (p \in X) \implies |f \circ \phi^{-1}| \text{ has a local maximum at } \phi(p_0).$$

$$|f \circ \phi^{-1}| \text{ has a local maximum} \xrightarrow{1.1.7} f|_U \text{ is constant} \xrightarrow{1.2.7} f \text{ is constant on } X.$$

This proves the compact maximum principle. $\square$

Definition 1.2.9 (Meromorphic functions). A function $f: W \setminus D \rightarrow \mathbb{C}$ is meromorphic on $W \subset X$ when some closed discrete $D \subset W$ satisfies

$$f|_{W \setminus D} \in \mathcal{O}_X(W \setminus D), \quad p \in D \implies p \text{ is a pole of } f,$$

$$\mathcal{M}_X(W) := \{f : f \text{ is meromorphic on } W\}.$$

Proposition 1.2.10 (Projective target). For a domain $W \subset X$ and $p_\infty = [1 : 0] \in \mathbb{P}^1$,

$$(1.2.1) \quad \mathcal{M}_X(W) \longleftrightarrow \{F \in \text{Hom}_{\text{hol}}(W, \mathbb{P}^1) \mid F \not\equiv p_\infty\}.$$

The correspondence is encoded by

$$\begin{array}{ccccc}
 W \setminus D & \xrightarrow{f} & \mathbb{C} & \xrightarrow{a \mapsto [a:1]} & \mathbb{P}^1 \\
 \downarrow & & & & \parallel \\
 W & \xrightarrow{\tilde{f}} & & & \mathbb{P}^1.
 \end{array}$$

Near $p \in D$, shrink $U \ni p$ so that $h := 1/f$ is holomorphic:

$$\begin{array}{ccccc}
 U \setminus \{p\} & \xrightarrow{f} & \mathbb{C}^\times & \xrightarrow{a \mapsto [a:1]} & U_\infty \subset \mathbb{P}^1 \\
 \downarrow & & \downarrow a \mapsto a^{-1} & & \downarrow [a:b] \mapsto b/a \\
 U & \xrightarrow{h} & \mathbb{C} & \xlongequal{\quad} & \mathbb{C}.
 \end{array}$$

Proof. Extend a meromorphic function across its pole set by its value at infinity.

$$\tilde{f}(p) := \begin{cases} [f(p) : 1], & p \in W \setminus D, \\ [1 : 0], & p \in D. \end{cases}$$

$$p \in D \implies \exists U \ni p, \quad h := 1/f \in \mathcal{O}_X(U), \quad h(p) = 0 \implies \tilde{f}|_U = [1 : h].$$

$$F \not\equiv p_\infty \xrightarrow{1.2.7} D := F^{-1}(p_\infty) \text{ is closed discrete}, \quad F|_{W \setminus D} = [f : 1].$$

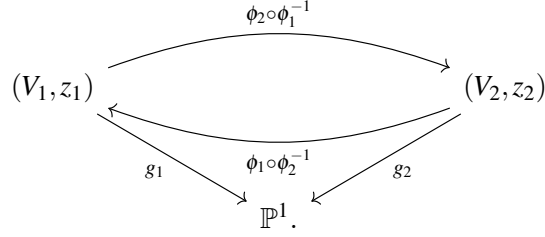
$$p \in D \implies F|_U = [1 : h], \quad h(p) = 0 \implies f|_{U \setminus D} = 1/h \text{ has a pole at } p.$$

Thus the two constructions are inverse. $\square$

Lemma 1.2.11 (Orders in charts). Let $0 \neq f \in \mathcal{M}_X(W)$, $p \in W$, and let

$$\phi_j: U_j \rightarrow V_j, \quad U_j \subset W, \quad z_j := \phi_j(p), \quad g_j := f \circ \phi_j^{-1}.$$

Then



If $m \in \mathbb{Z}$ and

$$g_1(z) = (z - z_1)^m \cdot u_1(z), \quad u_1(z_1) \neq 0,$$

then

$$\begin{aligned} \phi_1 \circ \phi_2^{-1}(w) - z_1 &= (w - z_2) \cdot a(w), \quad a(z_2) \neq 0, \\ g_2(w) &= (w - z_2)^m \cdot (a(w)^m \cdot u_1(\phi_1 \circ \phi_2^{-1}(w))). \end{aligned}$$

Thus the order and singularity type are chart-independent.

1.3. Plane Curves.

Definition 1.3.1 (Affine curves). Let $f \in \mathbb{C}[z, w]$ be nonconstant and squarefree, and put

$$X := V(f) \subset \mathbb{C}^2.$$

Then

$$\begin{aligned} \text{Sing}(X) &:= \left\{ p \in X : \begin{bmatrix} \left(\frac{\partial}{\partial z} f \right)(p) & \left(\frac{\partial}{\partial w} f \right)(p) \end{bmatrix} = [0 \ 0] \right\}, \\ X \text{ is smooth at } p &\iff p \notin \text{Sing}(X), \quad X \text{ is smooth} \iff \text{Sing}(X) = \emptyset. \end{aligned}$$

Theorem 1.3.2 (Holomorphic implicit function theorem). Let $f \in \mathcal{O}(U)$, $U \subset \mathbb{C}^2$, and $p = (z_0, w_0) \in V(f)$. After shrinking neighborhoods,

$$\begin{aligned} \left(\frac{\partial}{\partial w} f \right)(p) \neq 0 &\implies V(f) \cap U = \{(t, g(t)) : t \in V\}, \\ \left(\frac{\partial}{\partial z} f \right)(p) \neq 0 &\implies V(f) \cap U = \{(h(s), s) : s \in W\}. \end{aligned}$$

Equivalently,

$$\begin{array}{ccc} V & \xrightarrow{t \mapsto (t, g(t))} & V(f) \cap U \xrightarrow{\pi_z} V \\ & \searrow \text{id}_V & \nearrow \text{id}_W \\ W & \xrightarrow{s \mapsto (h(s), s)} & V(f) \cap U \xrightarrow{\pi_w} W. \end{array}$$

Theorem 1.3.3 (Smooth affine curves). Every connected component X_0 of a smooth affine plane curve is a noncompact Riemann surface.

Proof. Use the coordinate projection complementary to a nonzero partial derivative.

$$\left(\frac{\partial}{\partial w} f \right)(p) \neq 0 \xrightarrow{1.3.2} \phi_p(t, g_p(t)) = t, \quad \left(\frac{\partial}{\partial z} f \right)(q) \neq 0 \xrightarrow{1.3.2} \psi_q(h_q(s), s) = s.$$

$$\begin{array}{ccc} \phi_p(U_p \cap U_q) & \xrightarrow{\phi_q \circ \phi_p^{-1}} & \phi_q(U_p \cap U_q) \\ & \searrow \phi_p^{-1} & \downarrow \phi_q^{-1} \\ & & U_p \cap U_q \end{array} \quad \begin{array}{ccc} X_0 & \xrightarrow{(z, w)} & \mathbb{C}^2 \\ & \searrow z & \downarrow \pi_z \\ & & \mathbb{C}. \end{array}$$

$z|_{X_0}, w|_{X_0} \in \mathcal{O}_{X_0}(X_0) \wedge X_0 \text{ compact} \xrightarrow{1.2.8} z|_{X_0}, w|_{X_0} \text{ are constant} \implies X_0 \text{ is a point.}$
 $X_0 \text{ is locally biholomorphic to a nonempty open subset of } \mathbb{C} \implies X_0 \text{ is not a point.}$

Thus the graph charts give the asserted noncompact Riemann surface. $\square$

Definition 1.3.4 (Projective plane curves). Let

$$0 \neq F \in \mathbb{C}[x_0, x_1, x_2]$$

be squarefree and homogeneous of degree $d \geq 1$:

$$F(\lambda x_0, \lambda x_1, \lambda x_2) = \lambda^d \cdot F(x_0, x_1, x_2),$$

$$V(F) := \{[x_0 : x_1 : x_2] \in \mathbb{P}^2 : F(x_0, x_1, x_2) = 0\}.$$

Smoothness at $p \in V(F)$ is smoothness of any standard affine dehomogenization whose chart contains p .

Proposition 1.3.5 (Dehomogenization). Let

$$f(u, v) := F(1, u, v), \quad U_0 := \{x_0 \neq 0\}.$$

Then

$$(1.3.1) \quad F(x_0, x_1, x_2) = x_0^d \cdot f\left(\frac{x_1}{x_0}, \frac{x_2}{x_0}\right), \quad V(F) \cap U_0 \xrightarrow[\phi_0]{} V(f).$$

For $(u, v) = (x_1/x_0, x_2/x_0)$,

$$\left[\frac{\partial}{\partial x_0} F \quad \frac{\partial}{\partial x_1} F \quad \frac{\partial}{\partial x_2} F \right] = x_0^{d-1} \cdot \left[d \cdot f - u \cdot \frac{\partial}{\partial u} f - v \cdot \frac{\partial}{\partial v} f \quad \frac{\partial}{\partial u} f \quad \frac{\partial}{\partial v} f \right].$$

The affine and projective zero loci form

$$\begin{array}{ccc} V(F) \cap U_0 & \xhookrightarrow{\quad} & U_0 \\ \phi_0 \downarrow \sim & \searrow [x] \mapsto (x_1/x_0, x_2/x_0) \sim \downarrow \phi_0 & \\ V(f) & \xhookrightarrow{\quad} & \mathbb{C}^2. \end{array}$$

Lemma 1.3.6 (Euler identity). For homogeneous F of degree d ,

$$(1.3.2) \quad \left[\frac{\partial}{\partial x_0} F \quad \frac{\partial}{\partial x_1} F \quad \frac{\partial}{\partial x_2} F \right] \cdot \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix} = d \cdot F.$$

Indeed, for

$$m = c \cdot x_0^{a_0} x_1^{a_1} x_2^{a_2}, \quad a_0 + a_1 + a_2 = d,$$

$$\sum_{j=0}^2 x_j \cdot \frac{\partial}{\partial x_j} m = (a_0 + a_1 + a_2) \cdot m = d \cdot m.$$

Theorem 1.3.7 (Projective Jacobian criterion). For $X = V(F) \subset \mathbb{P}^2$ and $p \in X$,

$$p \in \text{Sing}(X) \iff \left[\left(\frac{\partial}{\partial x_0} F \right)(p) \quad \left(\frac{\partial}{\partial x_1} F \right)(p) \quad \left(\frac{\partial}{\partial x_2} F \right)(p) \right] = [0 \quad 0 \quad 0].$$

On U_0 , the criterion is transported by

$$\begin{array}{ccc} \text{Sing}(X) \cap U_0 & \xhookrightarrow{\quad} & X \cap U_0 \\ \phi_0 \downarrow \sim & \sim \downarrow \phi_0 & q = \left(\frac{x_1(p)}{x_0(p)}, \frac{x_2(p)}{x_0(p)} \right) \\ \text{Sing}(V(f)) & \xhookrightarrow{\quad} & V(f), \end{array}$$

Proof. Dehomogenize on a standard chart containing p .

$$\begin{aligned} \left[\left(\frac{\partial}{\partial x_1} F \right) (p) \quad \left(\frac{\partial}{\partial x_2} F \right) (p) \right] &= x_0(p)^{d-1} \cdot \left[\left(\frac{\partial}{\partial u} f \right) (q) \quad \left(\frac{\partial}{\partial v} f \right) (q) \right]. \\ x_0(p) \neq 0 \implies \left[\left(\frac{\partial}{\partial x_1} F \right) (p) = \left(\frac{\partial}{\partial x_2} F \right) (p) = 0 \iff \left(\frac{\partial}{\partial u} f \right) (q) = \left(\frac{\partial}{\partial v} f \right) (q) = 0 \right]. \\ F(p) = 0 \wedge \left(\frac{\partial}{\partial x_1} F \right) (p) = \left(\frac{\partial}{\partial x_2} F \right) (p) = 0 &\xrightarrow{1.3.2} x_0(p) \cdot \left(\frac{\partial}{\partial x_0} F \right) (p) = 0. \end{aligned}$$

Thus the affine criterion on the three standard charts gives the result. $\square$

Theorem 1.3.8 (Smooth projective curves). *Every smooth projective plane curve is a compact connected Riemann surface. For $x_k(p) \neq 0$, choose distinct j, k, ℓ with*

$$\left(\frac{\partial}{\partial x_j} F \right) (p) \neq 0.$$

Its local chart has the form

$$\begin{array}{ccccc} V & \xrightarrow{\Gamma} & V(F) \cap \mathcal{U}_p & \xrightarrow{\pi_\ell} & V \\ & \searrow & \downarrow \text{id}_V & \swarrow & \\ & & \mathcal{U}_p \subset U_k \cong \mathbb{C}^2. & & \end{array}$$

Proof. Put $X := V(F)$, fix $p \in X$, and combine compactness, the Jacobian criterion, and Bézout.

$$X \subset \mathbb{P}^2 \text{ closed} \wedge \mathbb{P}^2 \text{ compact} \implies X \text{ compact}.$$

$$x_k(p) \neq 0 \wedge p \notin \text{Sing}(X) \xrightarrow{1.3.2, 1.3.7} \exists j \neq k : \left(\frac{\partial}{\partial x_j} F \right) (p) \neq 0 \xrightarrow{1.3.2} X \cap U_k \text{ is locally a holomorphic graph.}$$

$$\begin{aligned} F &= G \cdot H, \quad G, H \text{ coprime of positive degree} \xrightarrow{A.0.2} \exists q \in V(G) \cap V(H). \\ \left(\frac{\partial}{\partial x_i} F \right) (q) &= G(q) \cdot \left(\frac{\partial}{\partial x_i} H \right) (q) + H(q) \cdot \left(\frac{\partial}{\partial x_i} G \right) (q) = 0 \quad (0 \leq i \leq 2) \implies q \in \text{Sing}(X). \\ X \text{ smooth} &\implies F \text{ irreducible} \implies X \text{ connected.} \end{aligned}$$

Thus the local graph atlas defines a compact connected Riemann surface. $\square$

Theorem 1.3.9 (Plane genus). *For a smooth plane curve $X \subset \mathbb{P}^2$ of degree d ,*

$$g(X) = \frac{(d-1)(d-2)}{2}.$$

The generic-projection computation is Proposition 2.3.5 in § 2.

Example 1.3.10 (Plane models). For

$$\begin{aligned} f(x, y) &= y^2 - x^2(x+1), \\ \left[f(x, y) \quad \left(\frac{\partial}{\partial x} f \right) (x, y) \quad \left(\frac{\partial}{\partial y} f \right) (x, y) \right] &= [0 \quad 0 \quad 0] \iff (x, y) = (0, 0). \\ y &= \pm x\sqrt{1+x} = \pm \left(x + \frac{1}{2}x^2 + O(x^3) \right), \end{aligned}$$

so $(0, 0)$ is a node. For

$$\begin{aligned} F(x, y, z) &:= y^2z - x^3 + xz^2, \\ z = 0 \wedge F = 0 &\implies [x : y : z] = [0 : 1 : 0], \quad \left(\frac{\partial}{\partial z} F \right) (0, 1, 0) = 1, \end{aligned}$$

$$z \neq 0, \quad [x : y : z] = [x : y : 1], \quad \frac{\partial}{\partial y} F = \frac{\partial}{\partial z} F = 0 \implies (x, y) = (0, 0),$$

$$\left(\frac{\partial}{\partial x} F \right) (0, 0, 1) = 1.$$

Hence $E := V(F)$ is smooth. For

$$Q(x, y, z) := x^4 + y^4 + z^4,$$

$$\left[\frac{\partial}{\partial x} Q \quad \frac{\partial}{\partial y} Q \quad \frac{\partial}{\partial z} Q \right] = 4 \cdot [x^3 \quad y^3 \quad z^3] = [0 \quad 0 \quad 0] \implies (x, y, z) = (0, 0, 0),$$

so the Fermat quartic is smooth. Hence

$$E, V(Q) \text{ smooth} \xrightarrow{1.3.9} g(E) = 1, \quad g(V(Q)) = 3.$$

2. MEROMORPHIC MAPS AND RAMIFICATION

Meromorphic germs are local fractions, while a nonconstant holomorphic map is controlled by its local multiplicities, covering degree, and ramification.

2.1. Orders and Meromorphic Functions.

Proposition 2.1.1 (Local fractions). *For an open set $W \subset X$,*

$$f \in \mathcal{M}_X(W) \iff \forall p \in W \quad f_p \in \text{Frac}(\mathcal{O}_{X,p}) \iff f|_U = \frac{g}{h},$$

where $U \ni p$ is a connected coordinate neighborhood, $g, h \in \mathcal{O}_X(U)$, and $h \neq 0$.

$$\begin{array}{ccc} U \setminus V(h) & \xrightarrow{(g,h)} & \mathbb{C} \times \mathbb{C}^\times \\ & \searrow f & \downarrow (a,b) \mapsto a/b \\ & & \mathbb{C} \end{array}$$

$$\begin{array}{ccccc} \mathcal{O}_X(U) & \hookrightarrow & \mathcal{M}_X(U) & & \\ \rho_{U,V} \downarrow & & \downarrow \rho_{U,V} & \searrow [\cdot]_p & \\ \mathcal{O}_X(V) & \hookrightarrow & \mathcal{M}_X(V) & & \\ & \searrow [\cdot]_p & \downarrow [\cdot]_p & \searrow [\cdot]_p & \\ & & \mathcal{O}_{X,p} & \hookrightarrow & \mathcal{M}_{X,p} \xrightarrow{\sim} \text{Frac}(\mathcal{O}_{X,p}). \end{array}$$

Definition 2.1.2 (Order). For $0 \neq f_p \in \mathcal{M}_{X,p}$, choose a coordinate z with $z(p) = 0$ and factor

$$f = z^r \cdot u, \quad r \in \mathbb{Z}, \quad u \in \mathcal{O}_{X,p}^\times, \quad \text{ord}_p(f) := r, \quad \text{ord}_p(0) := +\infty.$$

Then

$$\mathcal{O}_{X,p} = \{f \in \mathcal{M}_{X,p} : \text{ord}_p(f) \geq 0\}, \quad \mathfrak{m}_p = \{f \in \mathcal{M}_{X,p} : \text{ord}_p(f) > 0\},$$

$$\mathcal{O}_{X,p} \cong \mathbb{C}\{z\}, \quad \mathfrak{m}_p = z \cdot \mathcal{O}_{X,p}, \quad \mathcal{M}_{X,p} = \text{Frac}(\mathcal{O}_{X,p}).$$

Proposition 2.1.3 (Order arithmetic). *For $f, g \in \mathcal{M}_{X,p}^\times$,*

$$(2.1.1) \quad \text{ord}_p(fg) = \text{ord}_p(f) + \text{ord}_p(g), \quad \text{ord}_p\left(\frac{f}{g}\right) = \text{ord}_p(f) - \text{ord}_p(g).$$

$$\text{ord}_p(f+g) \geq \min\{\text{ord}_p(f), \text{ord}_p(g)\}.$$

Equality holds in the last relation when $\text{ord}_p(f) \neq \text{ord}_p(g)$.

Corollary 2.1.4 (Compact finiteness). *If X is compact and $0 \neq f \in \mathcal{M}_X(X)$, then*

$$\text{Supp}((f)_0) = \{p \in X : \text{ord}_p(f) > 0\}, \quad \text{Supp}((f)_\infty) = \{p \in X : \text{ord}_p(f) < 0\}$$

are finite.

Theorem 2.1.5 (Rational functions on the projective line). *For $t = x/y$ on $U_0 = \{y \neq 0\}$,*

$$\mathcal{M}_{\mathbb{P}^1}(\mathbb{P}^1) \xrightarrow{\sim} \mathbb{C}(t), \quad f \mapsto (t \mapsto f([t : 1])),$$

$$f([x : y]) = \frac{A(x, y)}{B(x, y)}, \quad A, B \in \mathbb{C}[x, y] \text{ coprime and homogeneous of one degree.}$$

$$\begin{array}{ccc} U_0 & \xrightarrow{f|_{U_0}} & \mathbb{P}^1 \\ \downarrow t \sim & \searrow & \downarrow f \\ \mathbb{C} & \xrightarrow{t \mapsto A(t, 1)/B(t, 1)} & \mathbb{P}^1. \end{array}$$

If $X = V(Q) \subset \mathbb{P}^2$ is smooth and G, H are homogeneous of one degree with $H|_X \neq 0$, then

$$\frac{G}{H} \in \mathcal{M}_X(X).$$

2.2. Multiplicity, Degree, and Fundamental Groups.

Definition 2.2.1 (Holomorphic maps). A map $F : X \rightarrow Y$ is holomorphic when its coordinate representative $f := \psi \circ F \circ \phi^{-1}$ is holomorphic.

$$\begin{array}{ccccc} & & G \circ F & & \\ & \xrightarrow{F} & & \xrightarrow{G} & \\ U_X & & U_Y & & U_Z \\ & \downarrow \phi \sim & \downarrow \psi \sim & \downarrow \sim \chi & \\ V_X & \xrightarrow{\psi \circ F \circ \phi^{-1}} & V_Y & \xrightarrow{\chi \circ G \circ \psi^{-1}} & V_Z \\ & \searrow \chi \circ G \circ F \circ \phi^{-1} & & & \end{array} \quad \begin{array}{ccc} X & \xrightarrow{F} Y & \xrightarrow{h} \mathbb{P}^1 \\ & \searrow h \circ F & \end{array}$$

Thus

$$G \circ F \text{ is holomorphic,} \quad F^*(h) := h \circ F \in \mathcal{M}_X(X) \quad (F \text{ nonconstant}).$$

Proposition 2.2.2 (Biholomorphisms). *For holomorphic maps of Riemann surfaces,*

$$X \xrightleftharpoons[F^{-1}]{F} Y \quad X \xrightarrow{F} Y \xrightarrow{G} Z.$$

$$F \text{ is a biholomorphism} \iff F \text{ is bijective,} \quad \text{mult}_p(G \circ F) = \text{mult}_p(F) \cdot \text{mult}_{F(p)}(G).$$

Theorem 2.2.3 (Local normal form). *Let $F : X \rightarrow Y$ be nonconstant and $p \in X$. There are centered coordinates for which*

$$(2.2.1) \quad \phi(p) = 0, \quad \psi(F(p)) = 0, \quad (\psi \circ F \circ \phi^{-1})(z) = z^{\text{mult}_p(F)}, \quad \text{mult}_p(F) \in \mathbb{Z}_{>0}.$$

$$\begin{array}{ccc} (U, p) & \xrightarrow{F} & (V, F(p)) \\ \downarrow \phi \sim & & \downarrow \psi \\ (\Delta, 0) & \xrightarrow{z \mapsto z^{\text{mult}_p(F)}} & (\Delta, 0) \end{array} \quad \begin{array}{ccc} T_p X & \xrightarrow{dF_p} & T_{F(p)} Y \\ \downarrow d\phi_p \sim & & \downarrow d\psi_{F(p)} \\ \mathbb{C} & \xrightarrow{\frac{\partial}{\partial z} z^{\text{mult}_p(F)}|_{z=0}} & \mathbb{C}. \end{array}$$

For a target parameter t at $F(p)$,

$$\text{mult}_p(F) = \text{ord}_p(t \circ F) = \dim_{\mathbb{C}} \left(\frac{\mathcal{O}_{X,p}}{(t \circ F) \cdot \mathcal{O}_{X,p}} \right).$$

Proof. Factor the first nonzero term of $g := \psi \circ F \circ \phi^{-1}$.

$$g(0) = 0, \quad g \not\equiv 0, \quad n := \min \left\{ k \geq 1 : \frac{d^k}{dz^k} g(z) \Big|_{z=0} \neq 0 \right\}, \quad g(z) = z^n \cdot H(z).$$

$$H(0) \neq 0 \implies H(z) = \exp(L(z)), \quad \eta(z) := \exp\left(\frac{L(z)}{n}\right), \quad \eta(z)^n = H(z).$$

$$\alpha(z) := z \cdot \eta(z), \quad \frac{d}{dz} \alpha(z) \Big|_{z=0} = \eta(0) \neq 0 \xrightarrow{1.1.8} \alpha \text{ is locally biholomorphic.}$$

$$g(z) = \alpha(z)^n \implies (\psi \circ F \circ (\alpha \circ \phi)^{-1})(w) = w^n, \quad n = \text{ord}_0(g) = \text{mult}_p(F).$$

This proves the local normal form and uniqueness of its exponent. $\square$

Definition 2.2.4 (Ramification). For nonconstant $F: X \rightarrow Y$, set

$$R := \{p \in X : \text{mult}_p(F) \geq 2\}, \quad B := F(R),$$

$$p \in R \iff \frac{\partial}{\partial z}(\psi \circ F \circ \phi^{-1})(z) \Big|_{z=0} = 0.$$

Theorem 2.2.5 (Finite maps and degree). Let $F: X \rightarrow Y$ be nonconstant and holomorphic, with X, Y compact and connected. Then F is open and surjective; its fibers, R , and B are finite.

$$\begin{array}{ccccc} R & \hookrightarrow & X & \hookleftarrow & X \setminus F^{-1}(B) \\ \downarrow F|_R & & & & \downarrow F^\circ \text{ deg}(F)\text{-sheeted} \\ B & \hookrightarrow & Y & \hookleftarrow & Y \setminus B \end{array} \quad \begin{array}{ccc} F^{-1}(q) & \hookrightarrow & X \\ \downarrow F|_{F^{-1}(q)} & & \downarrow F \\ \{q\} & \hookrightarrow & Y. \end{array}$$

The weighted fiber is independent of $q \in Y$:

$$(2.2.2) \quad \deg(F) := \sum_{p \in F^{-1}(q)} \text{mult}_p(F), \quad q \in Y.$$

Proposition 2.2.6 (Composition and principal divisors). Let $F: X \rightarrow Y$ and $G: Y \rightarrow Z$ be nonconstant holomorphic maps between compact connected Riemann surfaces. For $0 \neq f \in \mathcal{M}_X(X)$, let $\tilde{f}: X \rightarrow \mathbb{P}^1$ be the induced map.

$$\begin{array}{ccccc} X & \xrightarrow{F} & Y & \xrightarrow{G} & Z \\ & \searrow & \text{GoF} & \nearrow & \\ \tilde{f}^{-1}([0:1]) & \hookrightarrow & X & \hookleftarrow & \tilde{f}^{-1}([1:0]) \\ \downarrow & & \downarrow \tilde{f} & & \downarrow \\ \{[0:1]\} & \hookrightarrow & \mathbb{P}^1 & \hookleftarrow & \{[1:0]\} \end{array}$$

$$\text{mult}_p(G \circ F) = \text{mult}_p(F) \cdot \text{mult}_{F(p)}(G), \quad \deg(G \circ F) = \deg(G) \cdot \deg(F),$$

$$\deg(F) = 1 \iff F \text{ is a biholomorphism,} \quad \sum_{\text{ord}_p(f) > 0} \text{ord}_p(f) = \deg(\tilde{f}) = \sum_{\text{ord}_p(f) < 0} |\text{ord}_p(f)|.$$

Theorem 2.2.7 (Subgroup core and Galois closure). *Let $F : X \rightarrow Y$ be a nonconstant holomorphic map between compact connected Riemann surfaces, let B be its branch locus, and set*

$$Y^\circ := Y \setminus B, \quad X^\circ := X \setminus F^{-1}(B), \quad F^\circ := F|_{X^\circ}.$$

For $q_0 \in Y^\circ$, $p_0 \in (F^\circ)^{-1}(q_0)$, put

$$\Gamma := \pi_1(Y^\circ, q_0), \quad H := (F^\circ)_*(\pi_1(X^\circ, p_0)),$$

$$N := \text{Core}_\Gamma(H) := \bigcap_{\gamma \in \Gamma} \gamma \cdot H \cdot \gamma^{-1}.$$

Then $N \triangleleft \Gamma$, $N \subset H$, and the quotient tower compactifies uniquely to

$$\begin{array}{ccc} \tilde{Y}^\circ & \xrightarrow{/N} & \hat{X}^\circ \\ \searrow /H & & \downarrow \nu^\circ \\ & & X^\circ \\ \searrow / \Gamma & & \downarrow F^\circ \\ & & Y^\circ \end{array} \quad \begin{array}{ccc} \hat{X}^\circ & \hookrightarrow & \hat{X} \\ \downarrow \nu^\circ & & \downarrow \nu \\ \hat{X}^\circ & \hookrightarrow & X \\ \downarrow F^\circ & & \downarrow F \\ Y^\circ & \hookrightarrow & Y. \end{array}$$

Here $\hat{F} = F \circ \nu$ is Galois and

$$1 \longrightarrow N \longrightarrow \Gamma \twoheadrightarrow \text{Deck}(\hat{F}) \longrightarrow 1, \quad \text{Deck}(\hat{F}) \cong \Gamma/N.$$

Moreover,

$$\deg(F) = [\Gamma : H], \quad \deg(\hat{F}) = [\Gamma : N], \quad \deg(\nu) = [H : N],$$

$$F \text{ is Galois} \iff H \triangleleft \Gamma \iff N = H \iff |\text{Deck}(F)| = \deg(F).$$

Definition 2.2.8 (Orbifold fundamental group). Let Y have genus h , let $B = \{b_1, \dots, b_r\} \subset Y$, and let $m_j \geq 2$. For positive lassos γ_j about b_j , define

$$\Gamma_{Y, \mathbf{m}} := \pi_1^{\text{orb}}(Y; b_1/m_1, \dots, b_r/m_r) := \frac{\pi_1(Y \setminus B, q_0)}{\langle\langle \gamma_1^{m_1}, \dots, \gamma_r^{m_r} \rangle\rangle}.$$

Equivalently,

$$\Gamma_{Y, \mathbf{m}} = \left\langle \alpha_1, \beta_1, \dots, \alpha_h, \beta_h, \gamma_1, \dots, \gamma_r \left| \prod_{i=1}^h [\alpha_i, \beta_i] \cdot \prod_{j=1}^r \gamma_j = 1, \quad \gamma_j^{m_j} = 1 \right. \right\rangle,$$

$$\chi_{\text{orb}}(Y, \mathbf{m}) := 2 - 2h - \sum_{j=1}^r \left(1 - \frac{1}{m_j}\right).$$

Theorem 2.2.9 (Orbifold Riemann existence). *Let G be finite. The following are equivalent.*

(1) *There is a connected G -Galois holomorphic map*

$$F : X \longrightarrow Y$$

branched exactly over b_j , with ramification index m_j over b_j .

(2) *There are $A_i, B_i, C_j \in G$ satisfying*

$$G = \langle A_1, B_1, \dots, A_h, B_h, C_1, \dots, C_r \rangle,$$

$$\prod_{i=1}^h [A_i, B_i] \cdot \prod_{j=1}^r C_j = 1, \quad \text{ord}(C_j) = m_j.$$

(3) *There is an epimorphism*

$$\rho : \Gamma_{Y, \mathbf{m}} \twoheadrightarrow G$$

such that

$$\rho(\alpha_i) = A_i, \quad \rho(\beta_i) = B_i, \quad \rho(\gamma_j) = C_j, \quad \text{ord}(\rho(\gamma_j)) = m_j.$$

For the resulting cover,

$$\begin{array}{c} \begin{array}{ccc} & \tilde{Y}_{\text{orb}} & \\ \swarrow / \ker(\rho) & & \searrow / \Gamma_{Y, \mathbf{m}} \\ X & \xrightarrow{F} & Y \end{array} \\ 1 \longrightarrow \pi_1(X) \xrightarrow{\iota_F} \Gamma_{Y, \mathbf{m}} \xrightarrow{\rho} G \longrightarrow 1. \end{array}$$

Moreover,

$$G \cong \text{Deck}(F), \quad Y \cong X/G, \quad G_p := \{g \in G : g(p) = p\}, \quad G_p \cong \langle C_j \rangle, \quad |G_p| = m_j \quad (p \in F^{-1}(b_j)),$$

$$|F^{-1}(b_j)| = \frac{|G|}{m_j}, \quad 2 - 2g(X) = |G| \cdot \chi_{\text{orb}}(Y, \mathbf{m}).$$

2.3. Riemann–Hurwitz and Models.

Definition 2.3.1 (Ramification and branch divisors). Let $F : X \rightarrow Y$ be nonconstant and holomorphic between compact connected Riemann surfaces. For $q \in Y$, set

$$\begin{aligned} F^*(q) &:= \sum_{p \in F^{-1}(q)} \text{mult}_p(F) \cdot p, \quad \deg(F^*(q)) = \deg(F), \\ R_F &:= \sum_{p \in X} (\text{mult}_p(F) - 1) \cdot p, \quad B_F := F_*(R_F). \end{aligned}$$

Thus

$$\begin{aligned} B_F &= \sum_{q \in Y} \left(\sum_{p \in F^{-1}(q)} (\text{mult}_p(F) - 1) \right) \cdot q \\ &= \sum_{q \in Y} (\deg(F) - \#(F^{-1}(q))) \cdot q, \\ \text{Supp}(B_F) &= F(\text{Supp}(R_F)), \quad \deg(B_F) = \deg(R_F). \end{aligned}$$

Proposition 2.3.2 (Cotangent form of ramification). *The differential and its torsion cokernel $\mathcal{Q}_F$ fit into*

$$\begin{array}{ccccccc} 0 & \longrightarrow & F^* \Omega_Y & \xrightarrow{dF} & \Omega_X & \longrightarrow & \mathcal{Q}_F \longrightarrow 0 \\ & & \downarrow \bar{dF} \sim & & \parallel & & \downarrow \sim \\ 0 & \longrightarrow & \Omega_X(-R_F) & \longrightarrow & \Omega_X & \longrightarrow & \Omega_X / \Omega_X(-R_F) \longrightarrow 0. \end{array}$$

For every $p \in X$,

$$\begin{aligned} \text{length}_{\mathcal{O}_{X,p}}((\mathcal{Q}_F)_p) &= \text{mult}_p(F) - 1, \quad F^* \Omega_Y \cong \Omega_X(-R_F), \\ \Omega_X &\cong F^* \Omega_Y \otimes_{\mathcal{O}_X} \mathcal{O}_X(R_F), \quad K_X \sim F^* K_Y + R_F. \end{aligned}$$

Theorem 2.3.3 (Riemann–Hurwitz). *Let $F : X \rightarrow Y$ be as above. Then*

$$(2.3.1) \quad 2g(X) - 2 = \deg(F) \cdot (2g(Y) - 2) + \deg(R_F).$$

Proof. Remove the branch values $B = \{q_1, \dots, q_m\}$ and count punctures.

$$F^{-1}(q_j) = \{p_{j,1}, \dots, p_{j,r_j}\}, \quad n_{j,k} := \text{mult}_{p_{j,k}}(F), \quad \sum_{k=1}^{r_j} n_{j,k} = \deg(F).$$

$$\deg(R_F) = \sum_{j=1}^m \sum_{k=1}^{r_j} (n_{j,k} - 1) = m \cdot \deg(F) - \sum_{j=1}^m r_j.$$

$$\begin{array}{ccc} X \setminus F^{-1}(B) & \hookrightarrow & X \\ F^\circ \downarrow \text{deg}(F)\text{-sheeted} & & \downarrow F \\ Y \setminus B & \hookrightarrow & Y \end{array}$$

$$\chi(X) - \sum_{j=1}^m r_j = \chi(X \setminus F^{-1}(B)) = \deg(F) \cdot \chi(Y \setminus B) = \deg(F) \cdot (\chi(Y) - m).$$

$$\chi(X) = \deg(F) \cdot \chi(Y) - \deg(R_F) \stackrel{1.2.3}{\iff} 2g(X) - 2 = \deg(F) \cdot (2g(Y) - 2) + \deg(R_F).$$

This proves Equation (2.3.1). □

Corollary 2.3.4 (Genus constraints). *For $F: X \rightarrow \mathbb{P}^1$,*

$$\deg(R_F) = 2\deg(F) + 2g(X) - 2,$$

$$F: \mathbb{P}^1 \rightarrow \mathbb{P}^1, \quad \deg(F) = d \implies \deg(R_F) = 2d - 2.$$

For arbitrary Y ,

$$\begin{aligned} g(Y) \geq 1 &\implies g(X) \geq g(Y), & g(X) = 0 &\implies g(Y) = 0, \\ g(Y) = 1 &\implies g(X) \geq 1, & g(X) = g(Y) = 1 &\iff R_F = 0. \end{aligned}$$

Proposition 2.3.5 (Projection and plane genus). *Let $X = V(P) \subset \mathbb{P}^2$ be smooth of degree d , and choose $a = [0 : 0 : 1] \notin X$. Then*

$$\begin{array}{ccc} X & \hookrightarrow & \mathbb{P}^2 \setminus \{a\} \\ & \searrow \pi_a & \downarrow \pi_a \\ & & \mathbb{P}^1 \end{array} \quad \pi_a([x_0 : x_1 : x_2]) = [x_0 : x_1].$$

For $p \in X$,

$$\begin{aligned} T_p X &= V\left(\left(\frac{\partial}{\partial x_0} P\right)(p) \cdot x_0 + \left(\frac{\partial}{\partial x_1} P\right)(p) \cdot x_1 + \left(\frac{\partial}{\partial x_2} P\right)(p) \cdot x_2\right), \\ p \in \text{Supp}(R_{\pi_a}) &\iff a \in T_p X \iff \left(\frac{\partial}{\partial x_2} P\right)(p) = 0, \\ \text{mult}_p(\pi_a) - 1 &= I_p\left(X, V\left(\frac{\partial}{\partial x_2} P\right)\right). \end{aligned}$$

The polar and X have no common component, so

$$\deg(\pi_a) = d, \quad \deg(R_{\pi_a}) = d \cdot (d-1) \stackrel{2.3.3}{\implies} g(X) = \frac{(d-1)(d-2)}{2}.$$

For generic a , both R_{π_a} and B_{π_a} are reduced.

Example 2.3.6 (Hyperelliptic and cyclic covers). Let $h \in \mathbb{C}[x]$ be squarefree of degree b , and let X be the smooth projective model of $y^d = h(x)$.

$$\begin{array}{ccc}
V(y^d - h(x)) & \hookrightarrow & X \\
(x,y) \mapsto x \downarrow & & \downarrow \pi \\
\mathbb{C} & \xrightarrow{x \mapsto [x:1]} & \mathbb{P}^1.
\end{array}$$

If $d = 2$, then

$$\begin{aligned}
b = 2g + 1 &\implies \#(\text{Supp}(R_\pi)) = b + 1, & \infty \in \text{Supp}(R_\pi), \\
b = 2g + 2 &\implies \#(\text{Supp}(R_\pi)) = b, & \infty \notin \text{Supp}(R_\pi), \\
\deg(R_\pi) &= 2g + 2, & g(X) = g.
\end{aligned}$$

More generally,

$$\gcd(d, b) = 1 \implies \deg(\pi) = d, \quad \deg(R_\pi) = (b + 1) \cdot (d - 1), \quad g(X) = \frac{(d - 1)(b - 1)}{2}.$$

Example 2.3.7 (Maps between complex tori). Let $X = \mathbb{C}/L$ and $Y = \mathbb{C}/M$. Every holomorphic map lifts as

$$\begin{array}{ccc}
\mathbb{C} & \xrightarrow{z \mapsto \gamma \cdot z + a} & \mathbb{C} \\
\pi_L \downarrow & & \downarrow \pi_M \\
\mathbb{C}/L & \xrightarrow{F} & \mathbb{C}/M.
\end{array}$$

Indeed,

$$G(z + \lambda) - G(z) \in M \implies \left(\frac{\partial}{\partial z} G \right) (z + \lambda) = \left(\frac{\partial}{\partial z} G \right) (z) \xrightarrow{1.2.8} G(z) = \gamma \cdot z + a, \quad \gamma \cdot L \subset M.$$

Put

$$\phi_\gamma([z]_L) := [\gamma \cdot z]_M, \quad F = T_{[a]_M} \circ \phi_\gamma.$$

For $\gamma \neq 0$, F is unramified and

$$\begin{array}{ccccccc}
0 & \longrightarrow & L & \hookrightarrow & \gamma^{-1}M & \longrightarrow & \ker(\phi_\gamma) \longrightarrow 0 \\
& & \lambda \mapsto \gamma \lambda \downarrow \sim & & z \mapsto \gamma \cdot z \downarrow \sim & & \downarrow \sim \\
0 & \longrightarrow & \gamma L & \hookrightarrow & M & \longrightarrow & M/\gamma L \longrightarrow 0.
\end{array}$$

Hence

$$R_F = 0, \quad \deg(F) = [M : \gamma L] = |\gamma|^2 \cdot \frac{\text{covol}(L)}{\text{covol}(M)}.$$

Moreover,

$$\begin{aligned}
\text{Aut}(\mathbb{C}/L) &\cong (\mathbb{C}/L) \rtimes \text{Aut}_0(\mathbb{C}/L), & \text{Aut}_0(\mathbb{C}/L) &= \{\gamma \in \mathbb{C}^\times : \gamma \cdot L = L\}, \\
\text{Aut}_0(\mathbb{C}/L) &\cong \begin{cases} \mathbb{Z}/2\mathbb{Z}, & j(L) \neq 0, 1728, \\ \mathbb{Z}/4\mathbb{Z}, & j(L) = 1728, \\ \mathbb{Z}/6\mathbb{Z}, & j(L) = 0. \end{cases}
\end{aligned}$$

For $L_\tau := \mathbb{Z} + \mathbb{Z} \cdot \tau$, $\Im(\tau) > 0$,

$$\mathbb{C}/L_\tau \cong \mathbb{C}/L_{\tau'} \iff \begin{bmatrix} \tau' \\ 1 \end{bmatrix} = \frac{1}{c \cdot \tau + d} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \begin{bmatrix} \tau \\ 1 \end{bmatrix}, \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \text{SL}_2(\mathbb{Z}).$$

Proposition 2.3.8 (Finite quotient local model). *Let a finite group G act effectively and holomorphically on compact connected X . The quotient $Y := X/G$ has a unique Riemann-surface structure making $\pi: X \rightarrow Y$ holomorphic. For $g \in G$ and*

$$G_p := \{g \in G : g(p) = p\},$$

there is a G_p -invariant neighborhood $U \ni p$ for which

$$\begin{array}{ccc} X & \xrightarrow{g} & X \\ \pi \downarrow & & \downarrow \pi \\ X/G & \xrightarrow{1_{X/G}} & X/G \end{array} \quad \begin{array}{ccc} (U, p) & \xrightarrow[\sim]{\phi} & (\Delta, 0) \\ \pi \downarrow & & \downarrow z \mapsto z^{|G_p|} \\ (\pi(U), \pi(p)) & \xrightarrow[\sim]{\psi} & (\Delta, 0). \end{array}$$

The tangent representation gives

$$G_p \hookrightarrow \mathrm{GL}_{\mathbb{C}}(T_p X) \cong \mathbb{C}^\times \implies G_p \text{ is cyclic,} \\ \mathrm{mult}_p(\pi) = |G_p|, \quad \deg(\pi) = |G|.$$

Remark 2.3.9 (Nodal plane curves). If a reduced irreducible plane curve of degree d has exactly δ nodes and no other singularities, then its normalization has genus

$$g = \frac{(d-1)(d-2)}{2} - \delta.$$

3. MONODROMY, PROJECTIVE MODELS, AND DIFFERENTIALS

3.1. Monodromy and Hyperelliptic Covers. Let $F: X \rightarrow Y$ be a nonconstant holomorphic map of degree d between compact Riemann surfaces. Set

$$B := \{q \in Y : q \text{ is a branch value of } F\}, \quad Y^\circ := Y \setminus B, \quad X^\circ := X \setminus F^{-1}(B).$$

By Theorem 2.2.5,

B is finite, $F^\circ := F|_{X^\circ}: X^\circ \rightarrow Y^\circ$ is an unbranched degree- d covering, X°, Y° are connected.

$$\begin{array}{ccccc} (F^\circ)^{-1}(q_0) & \hookrightarrow & X^\circ & \hookrightarrow & X \\ \downarrow & & \downarrow F^\circ & & \downarrow F \\ \{q_0\} & \hookrightarrow & Y^\circ & \hookrightarrow & Y \end{array}$$

Definition 3.1.1 (Monodromy representation). Fix $q_0 \in Y^\circ$ and label

$$(F^\circ)^{-1}(q_0) = \{p_1, \dots, p_d\}.$$

For a loop $\gamma: [0, 1] \rightarrow Y^\circ$ based at q_0 , let $\tilde{\gamma}_i$ begin at p_i ; $\gamma \cdot \eta$ traverses η first.

$$\begin{array}{ccc} \{0\} & \xrightarrow{p_i} & X^\circ \\ \downarrow & \nearrow \tilde{\gamma}_i & \downarrow F^\circ \\ & & Y^\circ \\ & \xrightarrow{\gamma} & \end{array}$$

The monodromy permutation is determined by

$$\rho_F([\gamma])(i) = j \iff \tilde{\gamma}_i(1) = p_j, \quad \rho_F: \pi_1(Y^\circ, q_0) \rightarrow S_d.$$

For $\ell(p_i) = i$, the action is

$$\begin{array}{ccc}
\pi_1(Y^\circ, q_0) \times (F^\circ)^{-1}(q_0) & \xrightarrow{([\gamma], p_i) \mapsto \tilde{\gamma}(1)} & (F^\circ)^{-1}(q_0) \\
\rho_F \times \ell \downarrow & & \downarrow \ell \\
S_d \times \{1, \dots, d\} & \xrightarrow{(\sigma, i) \mapsto \sigma(i)} & \{1, \dots, d\}.
\end{array}$$

Proposition 3.1.2 (Well-definedness). *For the labeling above and a relabeling $p'_i = p_{\tau(i)}$,*

$$\rho_F([\gamma \cdot \eta]) = \rho_F([\gamma])\rho_F([\eta]), \quad \rho'_F([\gamma]) = \tau^{-1}\rho_F([\gamma])\tau.$$

Proof. Apply homotopy lifting and uniqueness for F° .

$$\begin{aligned}
\gamma_0 \simeq \gamma_1 \text{ rel } \{0, 1\} &\implies \tilde{\gamma}_{0,i}(1) = \tilde{\gamma}_{1,i}(1) \implies \rho_F([\gamma_0]) = \rho_F([\gamma_1]), \\
\widetilde{\gamma \cdot \eta}_i &= \tilde{\gamma}_{\rho_F([\eta])(i)} \cdot \tilde{\eta}_i \implies \rho_F([\gamma \cdot \eta]) = \rho_F([\gamma])\rho_F([\eta]), \\
\rho_F([\gamma^{-1}]) &= \rho_F([\gamma])^{-1}, \quad \rho_F([1_{q_0}]) = \text{id}, \\
p'_i = p_{\tau(i)} &\implies \rho'_F([\gamma]) = \tau^{-1}\rho_F([\gamma])\tau.
\end{aligned}$$

Thus monodromy is well defined up to conjugacy. $\square$

Proposition 3.1.3 (Components and orbits). *The components of X° are the monodromy orbits.*

$$\pi_0(X^\circ) \xrightarrow{\sim} \text{im}(\rho_F) \setminus \{1, \dots, d\}$$

In particular,

$$X^\circ \text{ is connected} \iff \text{im}(\rho_F) \text{ acts transitively on } \{1, \dots, d\}.$$

Proof. Fix $i, j \in \{1, \dots, d\}$.

$$\begin{aligned}
p_i, p_j \text{ lie in one component} &\implies \exists \tilde{\gamma}: p_i \rightsquigarrow p_j, \quad \gamma := F^\circ \circ \tilde{\gamma}, \\
\gamma(0) = \gamma(1) = q_0, \quad \tilde{\gamma}(0) = p_i, \quad \tilde{\gamma}(1) = p_j &\implies \rho_F([\gamma])(i) = j, \\
\rho_F([\gamma])(i) = j &\implies \tilde{\gamma}_i: p_i \rightsquigarrow p_j \implies p_i, p_j \text{ lie in one component,} \\
\pi_0(X^\circ) &\longleftrightarrow \text{im}(\rho_F) \setminus \{1, \dots, d\}.
\end{aligned}$$

Hence connectedness is equivalent to transitivity. $\square$

Theorem 3.1.4 (Classification by monodromy). *For a connected Riemann surface Y° ,*

$$\left\{ \begin{array}{l} \text{connected degree-}d \\ \text{coverings of } Y^\circ \end{array} \right\} / \cong \xrightarrow{\sim} \text{Hom}_{\text{trans}}(\pi_1(Y^\circ, q_0), S_d) / S_d.$$

Proof. Use the universal cover with its diagonal action.

$$\begin{array}{ccc}
G := \pi_1(Y^\circ, q_0), & (\tilde{q}, i) \cdot g := (\tilde{q} \cdot g, \rho(g)^{-1}(i)), & \\
X_\rho := (\tilde{Y}^\circ \times \{1, \dots, d\}) / G, & \tilde{Y}^\circ \times \{1, \dots, d\} \twoheadrightarrow X_\rho & \\
& \searrow (\tilde{q}, i) \mapsto \pi(\tilde{q}) \downarrow F_\rho & \\
& & Y^\circ
\end{array}$$

$$\begin{aligned}
\rho \text{ transitive} &\xleftrightarrow{3.1.3} X_\rho \text{ connected}, \quad \rho_{F_\rho} = \rho, \\
\rho' = \tau^{-1}\rho\tau &\iff [\tilde{q}, i] \mapsto [\tilde{q}, \tau^{-1}(i)] \text{ defines } X_\rho \xrightarrow{\sim} X_{\rho'}.
\end{aligned}$$

Thus the constructions are inverse modulo isomorphism and conjugacy. $\square$

Theorem 3.1.5 (Branch cycles on the sphere). *Fix $B = \{q_1, \dots, q_r\} \subset \mathbb{P}^1$. Let $\mathcal{H}_d(B)$ denote the connected degree- d covers branched only over B , and define*

$$(3.1.1) \quad \mathcal{T}_d(B) := \{(\sigma_1, \dots, \sigma_r) \in S_d^r : \sigma_1 \cdots \sigma_r = \text{id}, \langle \sigma_1, \dots, \sigma_r \rangle \text{ transitive}\},$$

$$\sigma_j \text{ has cycle lengths } e_{j1}, \dots, e_{jk_j} \implies \sum_{\ell=1}^{k_j} (e_{j\ell} - 1) = d - k_j.$$

All isomorphisms below are over $\mathbb{P}^1$. For positively oriented loops with $[\gamma_1] \cdots [\gamma_r] = 1$, set

$$\rho_F := (\rho_F([\gamma_1]), \dots, \rho_F([\gamma_r])).$$

Then

$$\mathcal{H}_d(B) / \cong \xrightarrow[\sim]{F \mapsto \rho_F} \mathcal{T}_d(B) / S_d.$$

The cycles of σ_j correspond to the points over q_j , with lengths equal to their ramification indices.

$$\begin{array}{ccccc} \bigsqcup_{\ell=1}^{k_j} D_{p_{j\ell}}^\times & \hookrightarrow & X^\circ & \hookrightarrow & X \\ \downarrow \bigsqcup_{\ell} z \mapsto z^{e_{j\ell}} & & \downarrow F^\circ & & \downarrow F \\ D_{q_j}^\times & \hookrightarrow & Y^\circ & \hookrightarrow & Y \end{array}$$

$$B \text{ is the branch locus} \iff \sigma_j \neq \text{id} \quad (1 \leq j \leq r).$$

Proof. Apply covering classification and complete across the punctures.

$$\begin{aligned} \pi_1(\mathbb{P}^1 \setminus B, q_0) &= \langle [\gamma_1], \dots, [\gamma_r] : [\gamma_1] \cdots [\gamma_r] = 1 \rangle, \\ \rho_F([\gamma_1]) \cdots \rho_F([\gamma_r]) &= \text{id}, \quad X^\circ \text{ connected} \xrightarrow{3.1.3} \langle \sigma_1, \dots, \sigma_r \rangle \text{ transitive}, \\ \sigma_j &= (c_{j1}) \cdots (c_{jk_j}), \quad |c_{j\ell}| = e_{j\ell} \xrightarrow{2.2.3} F \text{ has local model } z \mapsto z^{e_{j\ell}}, \\ \sum_{\ell=1}^{k_j} (\text{mult}_{p_{j\ell}}(F) - 1) &= \sum_{\ell=1}^{k_j} (e_{j\ell} - 1) = d - k_j. \end{aligned}$$

Thus the branch-cycle data determine the compact branched cover. □

Definition 3.1.6 (Hyperelliptic curve). A compact Riemann surface X is hyperelliptic when

$$\exists \pi : X \rightarrow \mathbb{P}^1 \text{ holomorphic}, \quad \deg(\pi) = 2.$$

Proposition 3.1.7 (Hyperelliptic involution). *Let $\pi : X \rightarrow \mathbb{P}^1$ have degree 2. The nontrivial permutation of a regular fiber extends to a holomorphic involution $\iota : X \rightarrow X$.*

$$\begin{array}{ccc} X & \xrightarrow{\iota} & X \\ \pi \downarrow & & \downarrow \pi \\ \mathbb{P}^1 & \xrightarrow{\text{id}} & \mathbb{P}^1 \end{array}$$

For the model $w^2 = h(z)$ from Example 2.3.6,

$$\begin{aligned} \iota(z, w) &= (z, -w), \quad \mathcal{M}_X(X) = \mathbb{C}(z) \oplus w\mathbb{C}(z), \\ r(z) &= \frac{f + f \circ \iota}{2}, \quad s(z) = \frac{f - f \circ \iota}{2w}, \quad f = r(z) + ws(z), \quad r, s \in \mathbb{C}(z), \\ \text{Fix}(\iota) &= \text{Ram}(\pi), \quad X/\langle \iota \rangle \cong \mathbb{P}^1, \quad \mathcal{M}_X(X)^{\langle \iota^* \rangle} = \mathbb{C}(z). \end{aligned}$$

3.2. Projective Curves and Models.

Definition 3.2.1 (Projective variety and dimension). A projective algebraic set has the form

$$X = V(F_1, \dots, F_m) \subset \mathbb{P}^n, \quad F_1, \dots, F_m \text{ homogeneous.}$$

It is a projective variety when it is irreducible, and

$$\dim(X) := \max \{r : X_0 \subsetneq X_1 \subsetneq \dots \subsetneq X_r = X, X_j \text{ irreducible}\}.$$

A projective curve is a projective variety of dimension 1.

Remark 3.2.2 (Noetherianity). Let $I \subset \mathbb{C}[x_0, \dots, x_n]$ be homogeneous. Hilbert's basis theorem gives

$$I = (F_1, \dots, F_m), \quad Z_1 \supseteq Z_2 \supseteq \dots \implies Z_N = Z_{N+1} = \dots, \\ V(I) = X_1 \cup \dots \cup X_s, \quad X_1, \dots, X_s \text{ irreducible.}$$

Theorem 3.2.3 (Projective dimension estimate). Let $F_1, \dots, F_r$ be homogeneous. Then

$$Z \text{ an irreducible component of } V(F_1, \dots, F_r) \implies \dim(Z) \geq n - r, \\ F_1, \dots, F_r \text{ regular} \implies V(F_1, \dots, F_r) \text{ is a pure complete intersection,} \\ \dim(V(F_1, \dots, F_r)) = n - r, \quad \dim(V(F)) = n - 1.$$

Example 3.2.4 (A non-pure zero locus). In $\mathbb{P}_{[x:y:z:w]}^3$,

$$V(xy, xz) = V(x) \cup V(y, z) \cong \mathbb{P}^2 \cup \mathbb{P}^1, \\ \dim(V(x)) = 2, \quad \dim(V(y, z)) = 1.$$

Definition 3.2.5 (Projective Jacobian). Let $X \subset \mathbb{P}^n$ be a projective curve with homogeneous ideal generated by $F_1, \dots, F_m$, and let $p = [p_0 : \dots : p_n] \in X$. For $\hat{p} = (p_0, \dots, p_n) \neq 0$, set

$$(3.2.1) \quad J_X(\hat{p}) := \begin{bmatrix} \left(\frac{\partial}{\partial x_0} F_1\right)(\hat{p}) & \left(\frac{\partial}{\partial x_1} F_1\right)(\hat{p}) & \dots & \left(\frac{\partial}{\partial x_n} F_1\right)(\hat{p}) \\ \left(\frac{\partial}{\partial x_0} F_2\right)(\hat{p}) & \left(\frac{\partial}{\partial x_1} F_2\right)(\hat{p}) & \dots & \left(\frac{\partial}{\partial x_n} F_2\right)(\hat{p}) \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{\partial}{\partial x_0} F_m\right)(\hat{p}) & \left(\frac{\partial}{\partial x_1} F_m\right)(\hat{p}) & \dots & \left(\frac{\partial}{\partial x_n} F_m\right)(\hat{p}) \end{bmatrix}, \\ p \in X_{\text{sm}} \iff \text{rank}(J_X(\hat{p})) = n - 1.$$

The rank condition is invariant under rescaling $\hat{p}$ and replacing the homogeneous generators.

Proposition 3.2.6 (Tangent space). For $p \in X_{\text{sm}}$,

$$T_p X \cong \ker(J_X(\hat{p}) : \mathbb{C}^{n+1} \rightarrow \mathbb{C}^m) / \mathbb{C}\hat{p}, \quad \dim_{\mathbb{C}}(T_p X) = 1.$$

The Euler identities give

$$J_X(\hat{p}) \cdot \hat{p}^T = \begin{bmatrix} \left(\frac{\partial}{\partial x_0} F_1\right)(\hat{p}) & \dots & \left(\frac{\partial}{\partial x_n} F_1\right)(\hat{p}) \\ \vdots & \ddots & \vdots \\ \left(\frac{\partial}{\partial x_0} F_m\right)(\hat{p}) & \dots & \left(\frac{\partial}{\partial x_n} F_m\right)(\hat{p}) \end{bmatrix} \cdot \begin{bmatrix} p_0 \\ \vdots \\ p_n \end{bmatrix} = \begin{bmatrix} \deg(F_1)F_1(\hat{p}) \\ \vdots \\ \deg(F_m)F_m(\hat{p}) \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Equivalently,

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathbb{C}\hat{p} & \hookrightarrow & \ker(J_X(\hat{p})) & \twoheadrightarrow & T_p X \longrightarrow 0 \\
& & \parallel & & \downarrow & & \downarrow \\
0 & \longrightarrow & \mathbb{C}\hat{p} & \hookrightarrow & \mathbb{C}^{n+1} & \twoheadrightarrow & \mathbb{C}^{n+1}/\mathbb{C}\hat{p} \longrightarrow 0 \\
& & & & \downarrow J_X(\hat{p}) & & \downarrow \bar{J}_X(\hat{p}) \\
& & & & \mathbb{C}^m & \xlongequal{\quad} & \mathbb{C}^m
\end{array}$$

Theorem 3.2.7 (Smooth projective curves). *Let $X \subset \mathbb{P}^n$ be connected and smooth. Then*

$$\text{rank}(J_X(\hat{p})) = n - 1 \xrightarrow{1.3.2} \dim_{\mathbb{C}}(T_p X) = 1 \implies X \text{ is a compact Riemann surface.}$$

Example 3.2.8 (The twisted cubic). The cubic Veronese map has image

$$\mathbb{P}^1 \xrightarrow{\sim} \mathbb{P}^3, \quad v_3([s:t]) = [s^3 : s^2 t : s t^2 : t^3] \hookrightarrow \mathbb{P}^3,$$

where

$$X = V(y^2 - xz, z^2 - yw, xw - yz) \subset \mathbb{P}^3_{[x:y:z:w]}.$$

The equations recover every point:

$$x \neq 0 \implies [x : y : z : w] = [1 : u : u^2 : u^3], \quad u := y/x, \quad x = 0 \implies [x : y : z : w] = [0 : 0 : 0 : 1].$$

Its projective Jacobian is

$$J_X(x, y, z, w) = \begin{bmatrix} \frac{\partial}{\partial x}(y^2 - xz) & \frac{\partial}{\partial y}(y^2 - xz) & \frac{\partial}{\partial z}(y^2 - xz) & \frac{\partial}{\partial w}(y^2 - xz) \\ \frac{\partial}{\partial x}(z^2 - yw) & \frac{\partial}{\partial y}(z^2 - yw) & \frac{\partial}{\partial z}(z^2 - yw) & \frac{\partial}{\partial w}(z^2 - yw) \\ \frac{\partial}{\partial x}(xw - yz) & \frac{\partial}{\partial y}(xw - yz) & \frac{\partial}{\partial z}(xw - yz) & \frac{\partial}{\partial w}(xw - yz) \end{bmatrix} = \begin{bmatrix} -z & 2y & -x & 0 \\ 0 & -w & 2z & -y \\ w & -z & -y & x \end{bmatrix}.$$

At $v_3([s : t])$, either $x = s^3 \neq 0$ or $w = t^3 \neq 0$, and

$$x \neq 0 \implies \det \begin{pmatrix} -x & 0 \\ -y & x \end{pmatrix} = -x^2 \neq 0, \quad w \neq 0 \implies \det \begin{pmatrix} 0 & -w \\ w & -z \end{pmatrix} = w^2 \neq 0.$$

$$z(y^2 - xz) + x(z^2 - yw) + y(xw - yz) = 0 \implies \text{rank}(J_X(p)) \leq 2 \quad (p \neq [0 : 0 : 0 : 1]),$$

$$J_X(0, 0, 0, 1) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \quad \text{rank}(J_X(p)) = 2 \quad (p \in X).$$

Hence

$$X \cong \mathbb{P}^1, \quad X \text{ is smooth and rational.}$$

Example 3.2.9 (Two crossing lines). Let $Z = V(xy, z) \subset \mathbb{P}^3$. Then

$$Z = V(x, z) \cup V(y, z), \quad J_Z(x, y, z, w) = \begin{bmatrix} \frac{\partial}{\partial x}(xy) & \frac{\partial}{\partial y}(xy) & \frac{\partial}{\partial z}(xy) & \frac{\partial}{\partial w}(xy) \\ \frac{\partial}{\partial x}z & \frac{\partial}{\partial y}z & \frac{\partial}{\partial z}z & \frac{\partial}{\partial w}z \end{bmatrix} = \begin{bmatrix} y & x & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Consequently,

$$\text{rank}(J_Z(x, y, z, w)) < 2 \iff x = y = z = 0 \iff [x : y : z : w] = [0 : 0 : 0 : 1],$$

$$\text{Sing}(Z) = \{[0 : 0 : 0 : 1]\} = V(x, z) \cap V(y, z).$$

Theorem 3.2.10 (Embedding in $\mathbb{P}^3$). *Let X be a smooth projective curve. Then*

$$X \cong X' \subset \mathbb{P}^3, \quad X' \text{ smooth and projective.}$$

Proof. Project successively from points outside the secant variety.

$$N \leq 3 \implies X \hookrightarrow \mathbb{P}^N \hookrightarrow \mathbb{P}^3, \quad N \geq 4 \implies \dim(\text{Sec}(X)) \leq 3 < N,$$

$$\dim(\text{Sec}(X)) < N \implies \exists q \in \mathbb{P}^N \setminus \text{Sec}(X), \quad \begin{array}{ccc} X & \hookrightarrow & \mathbb{P}^N \setminus \{q\} \\ \text{pr}_q|_X \downarrow \sim & & \downarrow \text{pr}_q \\ \text{pr}_q(X) & \hookrightarrow & \mathbb{P}^{N-1} \end{array}$$

$$q \notin \text{Sec}(X) \implies \text{pr}_q|_X \text{ is injective} \quad \wedge \quad d(\text{pr}_q|_X)_p \neq 0 \quad (p \in X),$$

$$X \text{ compact} \implies X \xrightarrow[\sim]{\text{pr}_q|_X} \text{pr}_q(X) \implies X \hookrightarrow \mathbb{P}^{N-1} \implies \dots \implies X \hookrightarrow \mathbb{P}^3.$$

Thus successive generic projections preserve X . □

Theorem 3.2.11 (Nodal plane model). *Every smooth projective curve X admits a finite birational normalization map onto a plane curve with only ordinary nodes.*

$$\begin{array}{ccc} X \setminus v^{-1}(\text{Sing}(C)) & \xrightarrow[\sim]{v} & C \setminus \text{Sing}(C) \\ \downarrow & & \downarrow \\ X & \xrightarrow{v} & C \subset \mathbb{P}^2. \end{array}$$

For a generic projection center q ,

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & \mathbb{P}^3 \setminus \{q\} \\ & \searrow v & \downarrow \text{pr}_q \\ & & C \subset \mathbb{P}^2 \end{array}$$

For every node $c \in C$,

$$v^{-1}(c) = \{p_c^+, p_c^-\}, \quad p_c^+, p_c^- \text{ are the two local branches.}$$

3.3. Differential Forms and Exterior Calculus.

Definition 3.3.1 (Holomorphic and meromorphic 1-forms). A holomorphic 1-form on X is a collection

$$\omega|_{U_i} = f_i(z_i) dz_i, \quad f_i \in \mathcal{O}(z_i(U_i)),$$

such that, for $T_{ij} := z_j \circ z_i^{-1}$,

$$f_i(z) = f_j(T_{ij}(z)) \frac{d}{dz} T_{ij}(z).$$

Equivalently,

$$\begin{array}{ccc} \mathcal{O}(z_j(U_i \cap U_j)) dz_j & \xrightarrow{T_{ij}^*} & \mathcal{O}(z_i(U_i \cap U_j)) dz_i \\ \uparrow & & \uparrow \\ f_j(z_j) dz_j & \longmapsto & f_j(T_{ij}(z_i)) \frac{d}{dz_i} T_{ij}(z_i) dz_i \end{array}$$

The holomorphic and meromorphic cotangent sheaves are

$$\Omega_X(U) = \{(f_i dz_i)_i : f_i dz_i = T_{ij}^*(f_j dz_j)\}, \quad \mathcal{M}\Omega_X := \Omega_X \otimes_{\mathcal{O}_X} \mathcal{M}_X.$$

Their gluing is the equalizer

$$\Omega_X(U) \hookrightarrow \prod_i \mathcal{O}(z_i(U_i)) dz_i \xrightarrow[T_{ij}^*]{\text{res}} \prod_{i,j} \mathcal{O}(z_i(U_i \cap U_j)) dz_i$$

At every $p \in X$,

$$\Omega_{X,p} \cong \mathcal{O}_{X,p} dz, \quad \text{rank}_{\mathcal{O}_{X,p}}(\Omega_{X,p}) = 1.$$

Definition 3.3.2 (Order and divisor of a differential). Let z be centered at p , and write

$$\omega = f(z) dz, \quad f(z) = z^m u(z), \quad m \in \mathbb{Z}, \quad u(0) \neq 0.$$

For a nonzero meromorphic 1-form on a compact Riemann surface,

$$\text{ord}_p(\omega) := m, \quad (\omega) := \sum_{p \in X} \text{ord}_p(\omega) p.$$

Lemma 3.3.3 (Coordinate invariance of order). For every centered coordinate at p ,

$$\text{ord}_p(\omega) \text{ is unchanged.}$$

Proof. Let $w = T(z)$ be a second centered coordinate.

$$\begin{aligned} \omega &= f(z) dz = g(w) dw, \quad f(z) = g(T(z)) \frac{d}{dz} T(z), \\ T(z) &= zH(z), \quad H(0) \neq 0, \quad g(w) = w^m G(w), \quad G(0) \neq 0, \\ f(z) &= z^m H(z)^m G(T(z)) \frac{d}{dz} T(z), \\ H(0)^m G(0) \frac{d}{dz} T(z) \Big|_{z=0} &\neq 0 \implies \text{ord}_0(f) = m = \text{ord}_0(g). \end{aligned}$$

Thus the order of a meromorphic differential is intrinsic. □

Definition 3.3.4 (Smooth forms). In a coordinate $z = x + iy$,

$$\eta = a(x, y) dx + b(x, y) dy = f(z, \bar{z}) dz + g(z, \bar{z}) d\bar{z}, \quad \Theta = h(x, y) dx \wedge dy.$$

The basis transformations are

$$\begin{aligned} \begin{bmatrix} dz \\ d\bar{z} \end{bmatrix} &= \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix} \cdot \begin{bmatrix} dx \\ dy \end{bmatrix}, \quad \begin{bmatrix} dx \\ dy \end{bmatrix} = \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \cdot \begin{bmatrix} dz \\ d\bar{z} \end{bmatrix}, \\ \begin{bmatrix} f \\ g \end{bmatrix} &= \frac{1}{2} \cdot \begin{bmatrix} 1 & -i \\ 1 & i \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix}, \quad dz \wedge d\bar{z} = -2i dx \wedge dy, \\ dx \wedge dy &= \frac{i}{2} dz \wedge d\bar{z}, \quad \mathcal{A}_X^1 = \mathcal{A}_X^{1,0} \oplus \mathcal{A}_X^{0,1}, \quad \mathcal{A}_X^2 = \mathcal{A}_X^{1,1}. \end{aligned}$$

Proposition 3.3.5 (Covector transformation). Let $w = u + iv = T(z)$ be a holomorphic coordinate change, where $z = x + iy$. Then

$$J_{\mathbb{R}}(T) = \begin{bmatrix} \frac{\partial}{\partial x} u & \frac{\partial}{\partial y} u \\ \frac{\partial}{\partial x} v & \frac{\partial}{\partial y} v \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} u & -\frac{\partial}{\partial x} v \\ \frac{\partial}{\partial x} v & \frac{\partial}{\partial x} u \end{bmatrix}.$$

For $\eta = A(w, \bar{w}) du + B(w, \bar{w}) dv$,

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} u & \frac{\partial}{\partial x} v \\ \frac{\partial}{\partial y} u & \frac{\partial}{\partial y} v \end{bmatrix} \cdot \begin{bmatrix} A \circ T \\ B \circ T \end{bmatrix} = J_{\mathbb{R}}(T)^{\top} \cdot \begin{bmatrix} A \circ T \\ B \circ T \end{bmatrix},$$

$$du \wedge dv = \det(J_{\mathbb{R}}(T)) dx \wedge dy = \left| \frac{d}{dz} T(z) \right|^2 dx \wedge dy.$$

Definition 3.3.6 (Exterior and Dolbeault differentials). For $f \in C_X^{\infty}$, set

$$(3.3.1) \quad \begin{aligned} df &= \frac{\partial}{\partial x} f dx + \frac{\partial}{\partial y} f dy = \frac{\partial}{\partial z} f dz + \frac{\partial}{\partial \bar{z}} f d\bar{z}, \\ \partial f &:= \frac{\partial}{\partial z} f dz, \quad \bar{\partial} f := \frac{\partial}{\partial \bar{z}} f d\bar{z}, \quad d = \partial + \bar{\partial}. \end{aligned}$$

For $\eta = f dz + g d\bar{z}$, define

$$(3.3.2) \quad \begin{aligned} \partial \eta &= \frac{\partial}{\partial z} g dz \wedge d\bar{z}, \quad \bar{\partial} \eta = -\frac{\partial}{\partial \bar{z}} f dz \wedge d\bar{z}, \\ d\eta &= \left(\frac{\partial}{\partial z} g - \frac{\partial}{\partial \bar{z}} f \right) dz \wedge d\bar{z}, \\ (f_1 dz + g_1 d\bar{z}) \wedge (f_2 dz + g_2 d\bar{z}) &= (f_1 g_2 - f_2 g_1) dz \wedge d\bar{z}. \end{aligned}$$

$$\begin{array}{ccccc} C_X^{\infty} & \xrightarrow{\partial} & \mathcal{A}_X^{1,0} & & \\ & \searrow d & \searrow & \searrow & \\ & & \mathcal{A}_X & \xrightarrow{d} & \mathcal{A}_X^2 \\ & \searrow \bar{\partial} & & \nearrow & \\ & & \mathcal{A}_X^{0,1} & & \end{array}$$

$$d(\alpha + \beta) = \bar{\partial} \alpha + \partial \beta, \quad \bar{\partial} \partial + \partial \bar{\partial} = 0.$$

The first-stage decomposition is

$$\begin{array}{ccc} C_X^{\infty} & \xrightarrow{(\partial, \bar{\partial})} & \mathcal{A}_X^{1,0} \oplus \mathcal{A}_X^{0,1} \\ d \downarrow & & \downarrow (\alpha, \beta) \mapsto \alpha + \beta \\ \mathcal{A}_X^1 & \xlongequal{\quad} & \mathcal{A}_X^1. \end{array}$$

The signed Dolbeault ladder is

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_X & \hookrightarrow & C_X^{\infty} & \xrightarrow{\bar{\partial}} & \mathcal{A}_X^{0,1} \longrightarrow 0 \\ & & \downarrow d & & \downarrow \partial & & \downarrow -\partial \\ 0 & \longrightarrow & \Omega_X & \hookrightarrow & \mathcal{A}_X^{1,0} & \xrightarrow{\bar{\partial}} & \mathcal{A}_X^{1,1} \longrightarrow 0 \end{array}$$

Proposition 3.3.7 (Coordinate invariance). For every holomorphic coordinate change T ,

$$\begin{aligned} T^* \circ d &= d \circ T^*, \quad T^* \circ \partial = \partial \circ T^*, \quad T^* \circ \bar{\partial} = \bar{\partial} \circ T^*, \\ T^*(\eta_1 \wedge \eta_2) &= T^* \eta_1 \wedge T^* \eta_2. \end{aligned}$$

Proof. Let $w = T(z)$ be a holomorphic coordinate change.

$$\begin{aligned}\varphi_i &= \varphi_j \circ T, & \eta_i &= a_i dz + b_i d\bar{z} = T^*(a_j dw + b_j d\bar{w}), \\ \begin{bmatrix} \frac{\partial}{\partial z} \varphi_i \\ \frac{\partial}{\partial \bar{z}} \varphi_i \end{bmatrix} &= \begin{bmatrix} \frac{d}{dz} T(z) & 0 \\ 0 & \frac{d}{d\bar{z}} T(z) \end{bmatrix} \cdot \begin{bmatrix} \left(\frac{\partial}{\partial w} \varphi_j \right) \circ T \\ \left(\frac{\partial}{\partial \bar{w}} \varphi_j \right) \circ T \end{bmatrix}, \\ \begin{bmatrix} a_i \\ b_i \end{bmatrix} &= \begin{bmatrix} \frac{d}{dz} T(z) & 0 \\ 0 & \frac{d}{d\bar{z}} T(z) \end{bmatrix} \cdot \begin{bmatrix} a_j \circ T \\ b_j \circ T \end{bmatrix}, & \begin{bmatrix} dw \\ d\bar{w} \end{bmatrix} &= \begin{bmatrix} \frac{d}{dz} T(z) & 0 \\ 0 & \frac{d}{d\bar{z}} T(z) \end{bmatrix} \cdot \begin{bmatrix} dz \\ d\bar{z} \end{bmatrix}, \\ \frac{\partial}{\partial z} b_i - \frac{\partial}{\partial \bar{z}} a_i &= \left| \frac{d}{dz} T(z) \right|^2 \left[\left(\frac{\partial}{\partial w} b_j - \frac{\partial}{\partial \bar{w}} a_j \right) \circ T \right], \\ dw \wedge d\bar{w} &= \left| \frac{d}{dz} T(z) \right|^2 dz \wedge d\bar{z}, & d\eta_i &= T^*(d\eta_j).\end{aligned}$$

Thus the local formulas define global operators. □

Lemma 3.3.8 (Nilpotence). *On smooth forms,*

$$d^2 = 0, \quad \partial^2 = 0, \quad \bar{\partial}^2 = 0, \quad \partial \bar{\partial} + \bar{\partial} \partial = 0.$$

Proof. Compute in a coordinate $z = x + iy$.

$$\begin{aligned}df &= \frac{\partial}{\partial x} f dx + \frac{\partial}{\partial y} f dy, \\ d(df) &= \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} f \right) dy \wedge dx + \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} f \right) dx \wedge dy, \\ d(df) &= \left[\frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} f \right) - \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} f \right) \right] dx \wedge dy = 0, \\ 0 &= d^2 f = \partial \bar{\partial} f + \bar{\partial} \partial f, & d: \mathcal{A}_X^2 &\longrightarrow \mathcal{A}_X^3 = 0.\end{aligned}$$

Hence the exterior derivative defines a complex. □

Theorem 3.3.9 (Poincaré lemma). *The smooth de Rham sequence is exact.*

$$0 \longrightarrow \mathbb{C} \longrightarrow C_X^\infty \xrightarrow{d} \mathcal{A}_X^1 \xrightarrow{d} \mathcal{A}_X^2 \longrightarrow 0$$

Proof. Work in a rectangular coordinate neighborhood.

$$\eta = a dx + b dy, \quad \Theta = h dx \wedge dy, \quad d\eta = 0 \iff \frac{\partial}{\partial x} b = \frac{\partial}{\partial y} a,$$

$$F(x, y) := \int_{x_0}^x a(t, y_0) dt + \int_{y_0}^y b(x, s) ds, \quad \begin{bmatrix} \frac{\partial}{\partial x} F \\ \frac{\partial}{\partial y} F \end{bmatrix} = \begin{bmatrix} a(x, y) \\ b(x, y) \end{bmatrix},$$

$$\eta = dF, \quad \beta := \left(\int_{x_0}^x h(t, y) dt \right) dy,$$

$$d\beta = \frac{\partial}{\partial x} \left(\int_{x_0}^x h(t, y) dt \right) dx \wedge dy = h(x, y) dx \wedge dy = \Theta.$$

Thus the de Rham complex is locally exact in positive degree. □

Remark 3.3.10 (Holomorphic de Rham complex). Local holomorphic integration gives

$$0 \longrightarrow \underline{\mathbb{C}} \longrightarrow \mathcal{O}_X \xrightarrow{d} \Omega_X \longrightarrow 0.$$

Its inclusion into the smooth resolution is

$$\begin{array}{ccccccc} 0 & \rightarrow & \underline{\mathbb{C}} & \rightarrow & \mathcal{O}_X & \xrightarrow{d} & \Omega_X \rightarrow 0 \\ \parallel & & \parallel & & \downarrow & & \downarrow \\ 0 & \rightarrow & \underline{\mathbb{C}} & \rightarrow & C_X^\infty & \xrightarrow{d} & \mathcal{A}_X^1 \xrightarrow{d} \mathcal{A}_X^2 \rightarrow 0 \end{array}$$

Therefore

$$H^k(X, \mathbb{C}) \cong \mathbb{H}^k\left(X, \mathcal{O}_X \xrightarrow{d} \Omega_X\right).$$

Definition 3.3.11 (Pullback). Let $F: X \rightarrow Y$ be holomorphic, with

$$w \circ F = H(z), \quad \eta = a(w, \bar{w}) dw + b(w, \bar{w}) d\bar{w}.$$

Then

$$F^* \eta = a(H(z), \overline{H(z)}) \frac{d}{dz} H(z) dz + b(H(z), \overline{H(z)}) \overline{\frac{d}{dz} H(z)} d\bar{z}.$$

At $p \in X$,

$$\begin{array}{ccc} T_{F(p)}^* Y \times T_p X & \xrightarrow{(dF_p)^* \times \text{id}} & T_p^* X \times T_p X \\ \text{id} \times dF_p \downarrow & & \downarrow \langle \cdot, \cdot \rangle \\ T_{F(p)}^* Y \times T_{F(p)} Y & \xrightarrow{\langle \cdot, \cdot \rangle} & \mathbb{C} \end{array}$$

Moreover,

$$F^{-1} \Omega_Y \xrightarrow{F^*} \Omega_X, \quad F^{-1} \mathcal{A}_Y^{p,q} \xrightarrow{F^*} \mathcal{A}_X^{p,q}, \quad F^{-1} \mathcal{M} \Omega_Y \xrightarrow{F^*} \mathcal{M} \Omega_X.$$

Proposition 3.3.12 (Functoriality of pullback). *For a holomorphic map $F: X \rightarrow Y$, the following squares commute.*

$$\begin{array}{ccc} C^\infty(Y) & \xrightarrow{d} & \mathcal{A}^1(Y) \\ F^* \downarrow & & \downarrow F^* \\ C^\infty(X) & \xrightarrow{d} & \mathcal{A}^1(X) \end{array} \quad \begin{array}{ccc} C^\infty(Y) & \xrightarrow{\partial} & \mathcal{A}^{1,0}(Y) \\ F^* \downarrow & & \downarrow F^* \\ C^\infty(X) & \xrightarrow{\partial} & \mathcal{A}^{1,0}(X) \end{array}$$

$$\begin{array}{ccc} C^\infty(Y) & \xrightarrow{\bar{\partial}} & \mathcal{A}^{0,1}(Y) \\ F^* \downarrow & & \downarrow F^* \\ C^\infty(X) & \xrightarrow{\bar{\partial}} & \mathcal{A}^{0,1}(X) \end{array} \quad \begin{array}{ccc} \mathcal{A}^1(Y) \times \mathcal{A}^1(Y) & \xrightarrow{\wedge} & \mathcal{A}^2(Y) \\ F^* \times F^* \downarrow & & \downarrow F^* \\ \mathcal{A}^1(X) \times \mathcal{A}^1(X) & \xrightarrow{\wedge} & \mathcal{A}^2(X) \end{array}$$

For $X \xrightarrow{F} Y \xrightarrow{G} Z$,

$$(G \circ F)^* = F^* \circ G^*, \quad \text{id}_X^* = \text{id}.$$

Corollary 3.3.13 (Holomorphic forms are closed). *If $\omega = f(z) dz \in \Omega_X(U)$, then*

$$d\omega = -\frac{\partial}{\partial \bar{z}} f dz \wedge d\bar{z} = 0.$$

Definition 3.3.14 (Contour integral). For a piecewise smooth path $\gamma: [a, b] \rightarrow X$,

$$\int_\gamma \eta := \int_a^b \gamma^* \eta.$$

If $z \circ \gamma = t \mapsto z(t)$ and $\eta = f dz + g d\bar{z}$, then

$$\int_{\gamma} \eta = \int_a^b \left[f(z(t), \overline{z(t)}) \frac{d}{dt} z(t) + g(z(t), \overline{z(t)}) \frac{d}{dt} \overline{z(t)} \right] dt.$$

Equivalently,

$$\begin{array}{ccc} \mathcal{A}^1(X) & \xrightarrow{\gamma^*} & \mathcal{A}^1([a, b]) \\ f_{\gamma} \downarrow & & \downarrow f_{[a, b]} \\ \mathbb{C} & \xlongequal{\quad} & \mathbb{C}. \end{array}$$

Theorem 3.3.15 (Stokes on a Riemann surface). *Let $M \subset X$ be a compact oriented subsurface with piecewise smooth boundary. Then*

$$\begin{array}{ccc} \mathcal{A}^1(M) & \xrightarrow{d} & \mathcal{A}^2(M) \\ \iota_{\partial M}^* \downarrow & & \downarrow f_M \\ \mathcal{A}^1(\partial M) & \xrightarrow{f_{\partial M}} & \mathbb{C} \end{array}$$

Theorem 3.3.16 (Stokes for paths). *For $f \in C^{\infty}(X)$ and $\gamma: [a, b] \rightarrow X$,*

$$\begin{array}{ccc} \int_{\gamma} df = f(\gamma(b)) - f(\gamma(a)). \\ C^{\infty}(X) & \xrightarrow{d} & \mathcal{A}^1(X) \\ \gamma^* \downarrow & & \downarrow \gamma^* \\ C^{\infty}([a, b]) & \xrightarrow{d} & \mathcal{A}^1([a, b]) \\ \text{ev}_b - \text{ev}_a \downarrow & & \downarrow f_{[a, b]} \\ \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

Proof. Apply the complex chain rule on coordinate subintervals.

$$\begin{aligned} \gamma^*(df) &= \left[\left(\frac{\partial}{\partial z} f \right) (z(t), \overline{z(t)}) \frac{d}{dt} z(t) + \left(\frac{\partial}{\partial \bar{z}} f \right) (z(t), \overline{z(t)}) \frac{d}{dt} \overline{z(t)} \right] dt, \\ \left(\frac{\partial}{\partial z} f \right) (z(t), \overline{z(t)}) \frac{d}{dt} z(t) + \left(\frac{\partial}{\partial \bar{z}} f \right) (z(t), \overline{z(t)}) \frac{d}{dt} \overline{z(t)} &= \frac{d}{dt} (f \circ \gamma)(t), \\ \int_{\gamma} df &= \int_a^b \frac{d}{dt} (f \circ \gamma)(t) dt, \\ \int_a^b \frac{d}{dt} (f \circ \gamma)(t) dt &= f(\gamma(b)) - f(\gamma(a)). \end{aligned}$$

Thus exact forms have zero integral around every closed path. □

4. RESIDUES, DIVISORS, AND THE CANONICAL CLASS

Throughout, X is a compact connected Riemann surface.

4.1. Residues.

Definition 4.1.1 (Residue of a meromorphic 1-form). Let $z(p) = 0$, let $\gamma_\varepsilon = \{|z| = \varepsilon\}$ have positive orientation, and write

$$\omega = \left(\sum_{n=-N}^{\infty} a_n z^n \right) dz.$$

Then

$$(4.1.1) \quad \text{Res}_p(\omega) := a_{-1} = \frac{1}{2\pi i} \int_{\gamma_\varepsilon} \omega.$$

Lemma 4.1.2 (Coordinate invariance). *For every local biholomorphism $T: (X, p) \rightarrow (Y, q)$,*

$$\begin{array}{ccc} \mathcal{M}\Omega_{Y,q} & \xrightarrow{\text{Res}_q} & \mathbb{C} \\ T^* \downarrow & & \parallel \\ \mathcal{M}\Omega_{X,p} & \xrightarrow{\text{Res}_p} & \mathbb{C}. \end{array}$$

In centered coordinates $w = T(z) = zu(z)$,

$$u(0) = \left. \frac{dw}{dz} T(z) \right|_{z=0} \neq 0, \quad T^* \left(\frac{dw}{w} \right) = \frac{dz}{z} + \frac{\frac{dw}{dz} u(z)}{u(z)} dz,$$

$$n \neq -1 \implies T^*(w^n dw) = \frac{1}{n+1} d(T(z)^{n+1}) \implies \text{Res}_p(T^* \omega) = \text{Res}_q(\omega).$$

Proposition 4.1.3 (Residue sequence). *Let $\Omega_X(p)$ consist of meromorphic 1-forms holomorphic away from p , with at most a simple pole at p . Then*

$$0 \longrightarrow \Omega_X \longrightarrow \Omega_X(p) \xrightarrow{\text{Res}_p} \mathbb{C}_p \longrightarrow 0.$$

For every coordinate $z(p) = 0$, the stalk sequence is

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Omega_{X,p} & \hookrightarrow & \Omega_X(p)_p & \xrightarrow{\text{Res}_p} & \mathbb{C} \longrightarrow 0 \\ & & \sim \downarrow & & \sim \downarrow & & \parallel \\ 0 & \longrightarrow & \mathcal{O}_{X,p} dz & \hookrightarrow & z^{-1} \mathcal{O}_{X,p} dz & \xrightarrow{\omega \mapsto d_{-1}\omega} & \mathbb{C} \longrightarrow 0. \end{array}$$

The sequence is exact as a sequence of sheaves; its map on global sections need not be surjective.

Proposition 4.1.4 (Logarithmic derivative). *For $f \in \mathcal{M}_{X,p}^\times$,*

$$f = z^m u, \quad u(0) \neq 0 \implies \frac{df}{f} = m \frac{dz}{z} + \frac{\frac{dw}{dz} u(z)}{u(z)} dz \implies \text{Res}_p \left(\frac{df}{f} \right) = \text{ord}_p(f).$$

$$\begin{array}{ccc} \mathcal{M}_{X,p}^\times & \xrightarrow{f \mapsto df/f} & \mathcal{M}\Omega_{X,p} \\ \text{ord}_p \downarrow & & \downarrow \text{Res}_p \\ \mathbb{Z} & \hookrightarrow & \mathbb{C}. \end{array}$$

Theorem 4.1.5 (Global residue theorem). *For every $\omega \in H^0(X, \mathcal{M}\Omega_X)$,*

$$\sum_{p \in X} \text{Res}_p(\omega) = 0.$$

$$\begin{array}{ccc}
H^0(X, \mathcal{M}\Omega_X) & \xrightarrow{(\text{Res}_p)_{p \in X}} & \bigoplus_{p \in X} \mathbb{C} \\
& \searrow 0 & \downarrow (c_p) \mapsto \sum_p c_p \\
& & \mathbb{C}.
\end{array}$$

If $p_1, \dots, p_r$ are the poles and $M = X \setminus \bigcup_{j=1}^r D_j^\circ$, then

$$\begin{aligned}
\partial M &= -\gamma_1 - \dots - \gamma_r, & d\omega &= 0, \\
0 &= \int_M d\omega \stackrel{3.3.15}{=} \int_{\partial M} \omega = -2\pi i \sum_{p \in X} \text{Res}_p(\omega).
\end{aligned}$$

The logarithmic derivative gives the global compatibility diagram

$$\begin{array}{ccc}
\mathcal{M}_X(X)^\times & \xrightarrow{f \mapsto df/f} & H^0(X, \mathcal{M}\Omega_X) \\
\downarrow \text{div} & & \downarrow (\text{Res}_p)_{p \in X} \\
\text{Div}(X) & \xrightarrow{\sum_p a_p p \mapsto (a_p)_p} & \bigoplus_{p \in X} \mathbb{C} \\
\downarrow \text{deg} & & \downarrow (c_p)_p \mapsto \sum_p c_p \\
\mathbb{Z} & \hookrightarrow & \mathbb{C}.
\end{array}$$

4.2. Divisor Classes and Line Bundles.

Definition 4.2.1 (Divisors).

$$\text{Div}(X) := \bigoplus_{p \in X} \mathbb{Z} \cdot p, \quad D = \sum_{p \in X} a_p p, \quad \text{Supp}(D) := \{p \in X : a_p \neq 0\}, \quad \text{deg}(D) := \sum_{p \in X} a_p.$$

The sum is finite, and $D \geq 0 \iff a_p \geq 0$ for every p . For $f \in \mathcal{M}_X(X)^\times$,

$$\text{div}(f) := \sum_{p \in X} \text{ord}_p(f) p, \quad \text{Prin}(X) := \{\text{div}(f) : f \in \mathcal{M}_X(X)^\times\}.$$

Definition 4.2.2 (Linear equivalence and the Picard group).

$$D \sim D' \iff D' - D = \text{div}(f) \quad \text{for some } f \in \mathcal{M}_X(X)^\times,$$

$$\text{Pic}(X) := \text{Div}(X) / \text{Prin}(X), \quad \text{Pic}^d(X) := \{[D] \in \text{Pic}(X) : \text{deg}(D) = d\}.$$

$$1 \longrightarrow \mathbb{C}^\times \longrightarrow \mathcal{M}_X(X)^\times \xrightarrow{\text{div}} \text{Div}(X) \longrightarrow \text{Pic}(X) \longrightarrow 0$$

$$\begin{array}{ccc}
\text{Div}(X) & \twoheadrightarrow & \text{Pic}(X) \\
\downarrow \text{deg} & & \downarrow \text{deg} \\
\mathbb{Z} & \xlongequal{\quad} & \mathbb{Z}.
\end{array}$$

Definition 4.2.3 (The divisor sheaf). For $D = \sum_p a_p p$,

$$\mathcal{O}_X(D)(U) := \{0\} \cup \{f \in \mathcal{M}_X(U)^\times : \text{ord}_p(f) + a_p \geq 0 \text{ for every } p \in U\}.$$

$$D \leq E \implies \mathcal{O}_X(D) \subset \mathcal{O}_X(E), \quad \mathcal{O}_X(-p) \subset \mathcal{O}_X \subset \mathcal{O}_X(p) \subset \mathcal{O}_X(2p) \subset \dots$$

Lemma 4.2.4 (Local triviality). *For every $D \in \text{Div}(X)$, the sheaf $\mathcal{O}_X(D)$ is locally free of rank 1. If*

$$D|_U = ap, \quad z(p) = 0,$$

then

$$\begin{aligned} \mathcal{O}_X(D)|_U &= z^{-a} \mathcal{O}_U, \quad f \in \mathcal{O}_X(D)(V) \iff z^a f \in \mathcal{O}_X(V). \\ \mathcal{O}_U &\xrightleftharpoons[f \mapsto z^a f]{h \mapsto z^{-a} h} \mathcal{O}_X(D)|_U \end{aligned}$$

Proposition 4.2.5 (Divisors and line bundles). *If $D' = D + \text{div}(g)$, then*

$$\begin{array}{ccc} \mathcal{O}_X(D) & \xrightarrow[\sim]{h \mapsto h/g} & \mathcal{O}_X(D') \\ \downarrow & & \downarrow \\ \mathcal{M}_X & \xrightarrow{h \mapsto h/g} & \mathcal{M}_X. \end{array}$$

Furthermore,

$$\mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(E) \cong \mathcal{O}_X(D + E), \quad \mathcal{O}_X(D)^\vee \cong \mathcal{O}_X(-D),$$

and every holomorphic line bundle on X has a nonzero meromorphic section. Hence

$$\begin{array}{ccc} \text{Pic}(X) & \xrightarrow{\sim} & \{\text{holomorphic line bundles on } X\} / \cong \\ \uparrow & & \uparrow \\ [D] & \longmapsto & \mathcal{O}_X(D). \end{array}$$

$$\mathcal{O}_X(D) \cong \mathcal{O}_X(E) \iff D \sim E.$$

Proposition 4.2.6 (The Picard group of $\mathbb{P}^1$). *For every $\infty \in \mathbb{P}^1$,*

$$\deg: \text{Pic}(\mathbb{P}^1) \xrightarrow{\sim} \mathbb{Z}, \quad [D] \mapsto \deg(D), \quad \mathcal{O}_{\mathbb{P}^1}(D) \cong \mathcal{O}_{\mathbb{P}^1}(\deg(D) \cdot \infty).$$

For $D = \sum_p a_p p$,

$$f_D(z) := \prod_{p \neq \infty} (z - z(p))^{a_p}, \quad \text{div}(f_D) = D - \deg(D) \cdot \infty.$$

$$\text{Pic}(\mathbb{P}^1) \xrightleftharpoons[n \mapsto [n\infty]]{\deg} \mathbb{Z}$$

4.3. Canonical Divisors and Pullback.

Definition 4.3.1 (Canonical class). A divisor K_X is canonical when

$$\mathcal{O}_X(K_X) \cong \Omega_X.$$

Equivalently, for every $0 \neq \omega \in H^0(X, \mathcal{M}\Omega_X)$,

$$K_X \sim \text{div}(\omega), \quad [K_X] \in \text{Pic}(X).$$

Lemma 4.3.2 (The canonical line bundle). *If $0 \neq \omega \in H^0(X, \mathcal{M}\Omega_X)$ and $K_X = \text{div}(\omega)$, then*

$$\begin{array}{ccc} \mathcal{O}_X(K_X) & \xrightarrow{\sim} & \Omega_X \\ \uparrow & & \uparrow \\ h & \longmapsto & h\omega. \end{array}$$

For nonzero meromorphic 1-forms η, ω ,

$$\begin{array}{ccc}
(H^0(X, \mathcal{M}\Omega_X) \setminus \{0\})^2 & \xrightarrow{(\eta, \omega) \mapsto \eta/\omega} & \mathcal{M}_X(X)^\times \\
(\eta, \omega) \mapsto ((\eta), (\omega)) \downarrow & & \downarrow f \mapsto \text{div}(f) \\
\text{Div}(X)^2 & \xrightarrow{(D, E) \mapsto D-E} & \text{Div}(X).
\end{array}$$

Thus

$$\eta = f\omega \implies \text{div}(\eta) - \text{div}(\omega) = \text{div}(f) \implies \text{div}(\eta) \sim \text{div}(\omega).$$

Example 4.3.3 (The canonical class of $\mathbb{P}^1$). For $w = 1/z$ at ∞ ,

$$dz = -\frac{1}{w^2} dw, \quad \text{div}(dz) = -2\infty, \quad K_{\mathbb{P}^1} \sim -2\infty, \quad \deg(K_{\mathbb{P}^1}) = -2.$$

Definition 4.3.4 (Pullback of divisors). Let $F: X \rightarrow Y$ be nonconstant. For $D = \sum_{q \in Y} a_q q$,

$$F^*(q) := \sum_{p \in F^{-1}(q)} \text{mult}_p(F) p, \quad F^*(D) := \sum_{q \in Y} a_q F^*(q).$$

Lemma 4.3.5 (Pullback laws). *Pullback is additive, and*

$$\begin{array}{ccc}
\mathcal{M}_Y(Y)^\times & \xrightarrow{\text{div}} & \text{Div}(Y) \\
g \mapsto g \circ F \downarrow & & \downarrow F^* \\
\mathcal{M}_X(X)^\times & \xrightarrow{\text{div}} & \text{Div}(X).
\end{array}$$

Therefore

$$\begin{array}{ccc}
\text{Div}(Y) & \xrightarrow{F^*} & \text{Div}(X) \\
\downarrow & & \downarrow \\
\text{Pic}(Y) & \xrightarrow{F^*} & \text{Pic}(X) \\
\downarrow \text{deg} & & \downarrow \text{deg} \\
\mathbb{Z} & \xrightarrow{n \mapsto \text{deg}(F) \cdot n} & \mathbb{Z}. \\
F^* \mathcal{O}_Y(D) & \cong & \mathcal{O}_X(F^*(D)).
\end{array}$$

Lemma 4.3.6 (Pullback order). *Let $F: X \rightarrow Y$ be nonconstant and holomorphic, let $p \in X$, let $q = F(p)$, and let $0 \neq \omega \in \mathcal{M}\Omega_{Y,q}$. Then*

$$\text{ord}_p(F^* \omega) = \text{mult}_p(F) (\text{ord}_q(\omega) + 1) - 1.$$

For $w = z^e$ and $\omega = w^m u(w) dw$,

$$e = \text{mult}_p(F), \quad F^* \omega = e z^{e(m+1)-1} u(z^e) dz, \quad m = \text{ord}_q(\omega).$$

Theorem 4.3.7 (Divisor form of Riemann–Hurwitz). *Let $F: X \rightarrow Y$ be nonconstant and holomorphic, and let R_F be its ramification divisor. Then*

$$0 \longrightarrow F^* \Omega_Y \xrightarrow{dF} \Omega_X \longrightarrow \Omega_{X/Y} \longrightarrow 0,$$

where $\Omega_{X/Y}$ is supported on $\text{Supp}(R_F)$. At $p \in X$, with $w = z^e$,

$$\begin{array}{ccccccc}
0 & \longrightarrow & (F^*\Omega_Y)_p & \xrightarrow{dF} & \Omega_{X,p} & \longrightarrow & \Omega_{X/Y,p} \longrightarrow 0 \\
& & \sim \downarrow & & \sim \downarrow & & \sim \downarrow \\
0 & \longrightarrow & \mathcal{O}_{X,p} & \xrightarrow{a \mapsto eze^{-1}a} & \mathcal{O}_{X,p} & \longrightarrow & \mathcal{O}_{X,p}/(z^{e-1}) \longrightarrow 0. \\
& & e = \text{mult}_p(F), & & \text{length}_{\mathcal{O}_{X,p}}(\Omega_{X/Y,p}) = e-1, & & \\
& & dF \in H^0(X, \Omega_X \otimes_{\mathcal{O}_X} (F^*\Omega_Y)^\vee), & & \text{div}(dF) = R_F. & &
\end{array}$$

Consequently,

$$\begin{array}{ccc}
(4.3.1) & \Omega_X \cong F^*\Omega_Y \otimes_{\mathcal{O}_X} \mathcal{O}_X(R_F), & K_X \sim F^*K_Y + R_F. \\
& F^*\Omega_Y \otimes_{\mathcal{O}_X} \mathcal{O}_X(R_F) & \xrightarrow{\sim} \Omega_X \\
& \sim \downarrow & \downarrow \sim \\
& \mathcal{O}_X(F^*K_Y + R_F) & \xrightarrow{\sim} \mathcal{O}_X(K_X).
\end{array}$$

Theorem 4.3.8 (Poincaré–Lelong). *For $0 \neq f \in \mathcal{M}_X(X)$, define*

$$\delta_{\text{div}(f)} := \sum_{p \in X} \text{ord}_p(f) \delta_p.$$

Then

$$\begin{array}{ccc}
\frac{i}{2\pi} \partial \bar{\partial} (\log(|f|^2)) & = & \delta_{\text{div}(f)}. \\
\mathcal{M}_X(X)^\times & \xrightarrow{\text{div}} & \text{Div}(X) \\
f \mapsto \frac{i}{2\pi} \partial \bar{\partial} \log(|f|^2) \downarrow & & \downarrow D \mapsto [D] \\
\mathcal{D}'^{1,1}(X) & \xlongequal{\quad} & \mathcal{D}'^{1,1}(X).
\end{array}$$

Locally, if $f = z^m u$ and $u(0) \neq 0$, then

$$\log(|f|^2) = m \log(|z|^2) + \log(|u|^2), \quad \partial \bar{\partial} (\log(|u|^2)) = 0.$$

For $\varphi \in C_c^\infty(\mathbb{C})$,

$$\begin{aligned}
\Delta_{\mathbb{R}}(\varphi) &:= \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \varphi \right) + \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \varphi \right), & \frac{i}{2} \partial \bar{\partial} \varphi &= \frac{1}{4} \Delta_{\mathbb{R}}(\varphi) dx \wedge dy, \\
\Delta_{\mathbb{R}}(\log(|z|^2)) &= 4\pi \delta_0, & \frac{i}{2\pi} \partial \bar{\partial} (\log(|z|^2)) &= \delta_0.
\end{aligned}$$

5. RIEMANN–ROCH, CANONICAL MAPS, AND JACOBIANS

Throughout, X is a compact connected Riemann surface of genus g , and $D \in \text{Div}(X)$.

5.1. Sheaf Cohomology and Riemann–Roch.

Definition 5.1.1 (Sheaves). A presheaf $\mathcal{F}$ assigns restriction homomorphisms $\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ for $V \subset U$, with

$$\rho_U^U = \text{id}_{\mathcal{F}(U)}, \quad W \subset V \subset U \implies \rho_W^U = \rho_W^V \circ \rho_V^U.$$

It is a sheaf precisely when every open cover $U = \bigcup_i U_i$ gives the exact equalizer

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j).$$

The principal examples are

$$\mathcal{O}_X, \quad \mathcal{M}_X, \quad \Omega_X, \quad \mathcal{A}_X^{p,q}, \quad \mathcal{O}_X(D).$$

Definition 5.1.2 (Sheaf cohomology). By Proposition ??, every sheaf $\mathcal{F}$ of $\mathcal{O}_X$ -modules admits

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}^0 \xrightarrow{\delta^0} \mathcal{I}^1 \xrightarrow{\delta^1} \mathcal{I}^2 \longrightarrow \dots,$$

and global sections give

$$0 \longrightarrow \mathcal{I}^0(X) \xrightarrow{\delta_X^0} \mathcal{I}^1(X) \xrightarrow{\delta_X^1} \mathcal{I}^2(X) \longrightarrow \dots.$$

Define

$$H^0(X, \mathcal{F}) := \mathcal{F}(X), \quad H^q(X, \mathcal{F}) := \frac{\ker(\delta_X^q)}{\operatorname{im}(\delta_X^{q-1})} \quad (q \geq 1),$$

$$H^q(D) := H^q(X, \mathcal{O}_X(D)), \quad h^q(D) := \dim_{\mathbb{C}}(H^q(D)).$$

Every short exact sequence $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ gives, by Theorem ??,

$$\begin{array}{ccccccc} 0 & \rightarrow & H^0(X, \mathcal{F}') & \rightarrow & H^0(X, \mathcal{F}) & \rightarrow & H^0(X, \mathcal{F}'') \\ & & \swarrow \delta & & & & \\ & & H^1(X, \mathcal{F}') & \rightarrow & H^1(X, \mathcal{F}) & \rightarrow & H^1(X, \mathcal{F}'') \rightarrow \dots \end{array}$$

Theorem 5.1.3 (Dolbeault resolution and Hodge decomposition). For every holomorphic line bundle $\mathcal{L}$,

$$0 \longrightarrow \mathcal{L} \hookrightarrow \mathcal{A}_X^{0,0}(\mathcal{L}) \xrightarrow{\bar{\partial}_{\mathcal{L}}} \mathcal{A}_X^{0,1}(\mathcal{L}) \longrightarrow 0.$$

Theorem 5.1.2 gives

$$0 \rightarrow H^0(X, \mathcal{L}) \hookrightarrow \mathcal{A}_X^{0,0}(X, \mathcal{L}) \xrightarrow{\bar{\partial}_{\mathcal{L}}} \mathcal{A}_X^{0,1}(X, \mathcal{L}) \twoheadrightarrow H^1(X, \mathcal{L}) \rightarrow 0,$$

and

$$H^1(X, \mathcal{L}) \cong \frac{\mathcal{A}_X^{0,1}(X, \mathcal{L})}{\operatorname{im}(\bar{\partial}_{\mathcal{L}}: \mathcal{A}_X^{0,0}(X, \mathcal{L}) \rightarrow \mathcal{A}_X^{0,1}(X, \mathcal{L}))}, \quad H^q(X, \mathcal{L}) = 0 \quad (q \geq 2).$$

For $\mathcal{L} = \mathcal{O}_X$,

$$\begin{array}{ccc} H_{\text{dR}}^1(X; \mathbb{C}) & \xrightarrow{\sim} & H_{\bar{\partial}}^{1,0}(X) \oplus H_{\bar{\partial}}^{0,1}(X) \\ & & \downarrow \sim \\ & & H^0(X, \Omega_X) \oplus H^1(X, \mathcal{O}_X). \end{array}$$

Complex conjugation exchanges the two summands, and

$$\dim_{\mathbb{C}}(H_{\text{dR}}^1(X; \mathbb{C})) = 2g \xrightarrow{1.2.3} g = \dim_{\mathbb{C}}(H^0(X, \Omega_X)) = \dim_{\mathbb{C}}(H^1(X, \mathcal{O}_X)).$$

Definition 5.1.4 (Global sections and complete linear systems). Definition 4.2.3 gives

$$H^0(D) = \{0\} \cup \{f \in \mathcal{M}_X(X)^{\times} : \operatorname{div}(f) + D \geq 0\}, \quad |D| := \{E \in \operatorname{Div}(X) : E \geq 0 \wedge E \sim D\}.$$

Lemma 5.1.5 (Evaluation and dimension jump). Let $i_p: \{p\} \hookrightarrow X$, let $\mathcal{L} = \mathcal{O}_X(D)$, and put $\mathcal{L}_p := i_{p*}(\mathcal{L}|_p)$. In the cohomology row, identify $H^0(X, \mathcal{L}_p) \cong \mathcal{L}|_p$ and suppress $H^0(X, -)$. Then

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{O}_X(D-p) & \hookrightarrow & \mathcal{O}_X(D) & \longrightarrow & \mathcal{L}_p \longrightarrow 0, \\ 0 & \longrightarrow & H^0(D-p) & \hookrightarrow & H^0(D) & \xrightarrow{\operatorname{ev}_p} & \mathcal{L}_p \longrightarrow H^1(D-p). \end{array}$$

The latter continues as

$$H^1(D-p) \longrightarrow H^1(D) \longrightarrow 0.$$

If $D|_U = mp$ and $z(p) = 0$, then

$$\frac{\mathcal{O}_X(D)_p}{\mathcal{O}_X(D-p)_p} = \frac{z^{-m}\mathcal{O}_{X,p}}{z^{1-m}\mathcal{O}_{X,p}} \cong \mathcal{L}|_p \cong \mathbb{C}, \quad \ker(\text{ev}_p) = H^0(D-p),$$

$$0 \leq h^0(D) - h^0(D-p) \leq 1, \quad \deg(D) < 0 \xrightarrow{2.2.6} h^0(D) = 0,$$

$$N \in \mathbb{Z}_{>0}, \quad N > \deg(D) \implies \deg(D-Np) < 0 \xrightarrow{2.2.6} H^0(D-Np) = 0.$$

$$D_j := D - jp, \quad H^0(D_N) = 0 \implies h^0(D) = \sum_{j=0}^{N-1} (h^0(D_j) - h^0(D_{j+1})) \leq N < \infty.$$

Proposition 5.1.6 (Projectivization of sections). *By*

$$\begin{array}{ccc} \mathbb{P}H^0(D) & \xrightarrow{\sim} & |D| \\ \uparrow & & \uparrow \\ [f] & \longmapsto & D + \text{div}(f). \end{array}$$

Indeed,

$$D + \text{div}(f) = D + \text{div}(h) \iff \text{div}(f/h) = 0 \xrightarrow{1.2.8} f/h \in \mathbb{C}^\times \iff [f] = [h].$$

If $D' = D + \text{div}(g)$, then

$$H^0(D) \xrightarrow{\sim} H^0(D'), \quad f \mapsto \frac{f}{g}, \quad \text{div}(f/g) + D' = \text{div}(f) + D, \quad |D| = |D'|.$$

Theorem 5.1.7 (Serre duality). *For every holomorphic line bundle $\mathcal{L}$, integration induces*

$$H^1(X, \mathcal{L})^\vee \cong H^0(X, \Omega_X \otimes_{\mathcal{O}_X} \mathcal{L}^\vee).$$

For $\mathcal{L} = \mathcal{O}_X(D)$,

$$\begin{aligned} H^1(D) \otimes H^0(K_X - D) &\longrightarrow H^1(K_X) \xrightarrow{\text{Tr}_X} \mathbb{C}, \\ ([\alpha], s) &\longmapsto \int_X s \wedge \alpha, \quad H^1(D)^\vee \cong H^0(K_X - D), \quad h^1(D) = h^0(K_X - D). \end{aligned}$$

Equivalently,

$$\begin{array}{ccc} \text{Pic}^d(X) & \xrightarrow{\mathcal{L} \mapsto \Omega_X \otimes_{\mathcal{O}_X} \mathcal{L}^\vee} & \text{Pic}^{\deg(K_X)-d}(X) \\ \downarrow \mathcal{L} \mapsto h^1(X, \mathcal{L}) & & \downarrow \mathcal{M} \mapsto h^0(X, \mathcal{M}) \\ \mathbb{Z}_{\geq 0} & \xlongequal{\quad} & \mathbb{Z}_{\geq 0}. \end{array}$$

Theorem 5.1.8 (Riemann–Roch). *Let $\mathcal{L} \cong \mathcal{O}_X(D)$, and define $\deg(\mathcal{L}) := \deg(D)$. Then*

(5.1.1)

$$\chi(X, \mathcal{L}) := h^0(X, \mathcal{L}) - h^1(X, \mathcal{L}) = \deg(\mathcal{L}) - g + 1, \quad h^0(D) - h^0(K_X - D) = \deg(D) - g + 1.$$

$$\begin{array}{ccc} \text{Pic}(X) & \xrightarrow{\mathcal{L} \mapsto \chi(X, \mathcal{L})} & \mathbb{Z} \\ \deg \downarrow & & \parallel \\ \mathbb{Z} & \xrightarrow{d \mapsto d - g + 1} & \mathbb{Z}. \end{array}$$

Proof. Apply Euler characteristic to the point-extension sequence and then use Serre duality.

$$X \text{ compact and connected} \xrightarrow{1.2.8, 5.1.3} h^0(0) = 1, \quad h^1(0) = g, \quad \chi(X, \mathcal{O}_X) = 1 - g.$$

$$0 \rightarrow \mathcal{O}_X(D - p) \rightarrow \mathcal{O}_X(D) \rightarrow i_{p*}(\mathcal{O}_X(D)|_p) \rightarrow 0 \xrightarrow{??} \chi(X, \mathcal{O}_X(D)) - \chi(X, \mathcal{O}_X(D - p)) = 1.$$

$$D = \sum_{p \in X} a_p p \implies \chi(X, \mathcal{O}_X(D)) = \chi(X, \mathcal{O}_X) + \sum_{p \in X} a_p = \deg(D) - g + 1.$$

$$h^1(D) = h^0(K_X - D) \xrightarrow{5.1.7} h^0(D) - h^0(K_X - D) = \deg(D) - g + 1.$$

This proves Equation (5.1.1). $\square$

Corollary 5.1.9 (Riemann–Roch consequences).

$$D = K_X \xrightarrow{5.1.1} \deg(K_X) = 2g - 2, \quad h^0(D) = \deg(D) - g + 1 + h^0(K_X - D).$$

Define the index of speciality by

$$i(D) := h^1(D) = h^0(K_X - D), \quad D \text{ is special} \iff i(D) > 0.$$

Moreover,

$$\begin{aligned} h^0(D) &\geq \deg(D) - g + 1, \quad \deg(D) \geq 2g - 1 \implies \deg(K_X - D) < 0 \xrightarrow{5.1.5, 5.1.7} i(D) = 0, \\ \deg(D) &\geq 2g - 1 \implies h^0(D) = \deg(D) - g + 1. \end{aligned}$$

5.2. Linear Systems and Canonical Maps.

Definition 5.2.1 (Projective linear systems). Let $V \subset H^0(D)$ satisfy $\dim_{\mathbb{C}}(V) = n + 1$. Define

$$|V| := \{D + \operatorname{div}(s) : 0 \neq s \in V\} \subset |D|, \quad F_V := \min_{0 \neq s \in V} (D + \operatorname{div}(s)),$$

$$\operatorname{Bs}(|V|) := \operatorname{Supp}(F_V), \quad |V| \text{ is base-point free} \iff F_V = 0.$$

Equivalently,

$$V \otimes_{\mathbb{C}} \mathcal{O}_X \xrightarrow{\operatorname{ev}} \mathcal{O}_X(D).$$

Proposition 5.2.2 (Linear systems and projective maps). Let $V \subset H^0(D)$ be base-point free, with basis $s_0, \dots, s_n$. Then

$$\phi_V : X \longrightarrow \mathbb{P}(V^\vee) \cong \mathbb{P}^n, \quad p \longmapsto [\operatorname{ev}_p] = [s_0(p) : \dots : s_n(p)].$$

Every local frame of $\mathcal{O}_X(D)|_U$ gives

$$\begin{array}{ccc} U & \xrightarrow{p \mapsto (s_0(p), \dots, s_n(p))} & \mathbb{C}^{n+1} \setminus \{0\} \\ & \searrow \phi_V & \swarrow \\ & \mathbb{P}^n & \end{array}$$

The pulled-back universal quotient is

$$\begin{array}{ccc} \phi_V^*(V \otimes_{\mathbb{C}} \mathcal{O}_{\mathbb{P}(V^\vee)}) & \xrightarrow{\phi_V^*(q)} & \phi_V^* \mathcal{O}_{\mathbb{P}(V^\vee)}(1) \\ \sim \downarrow & & \downarrow \sim \\ V \otimes_{\mathbb{C}} \mathcal{O}_X & \xrightarrow{\operatorname{ev}} & \mathcal{O}_X(D). \end{array}$$

Conversely, every nondegenerate $\phi : X \rightarrow \mathbb{P}^n$ arises from

$$\mathcal{L} := \phi^* \mathcal{O}_{\mathbb{P}^n}(1) \cong \mathcal{O}_X(D), \quad V := \operatorname{im}(H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(1)) \rightarrow H^0(X, \mathcal{L})),$$

uniquely up to a linear change of homogeneous coordinates.

Proposition 5.2.3 (Hyperplane divisors). *For $\mathbf{a} = [a_0 : \cdots : a_n] \in (\mathbb{P}^n)^\vee$,*

$$H_{\mathbf{a}} = V \left(\begin{bmatrix} a_0 & \cdots & a_n \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ \vdots \\ x_n \end{bmatrix} \right), \quad s_{\mathbf{a}} := \begin{bmatrix} a_0 & \cdots & a_n \end{bmatrix} \cdot \begin{bmatrix} s_0 \\ \vdots \\ s_n \end{bmatrix}.$$

Then

$$\phi_V^*(H_{\mathbf{a}}) = D + \text{div}(s_{\mathbf{a}}), \quad |V| = \{\phi_V^*(H) : H \subset \mathbb{P}^n \text{ a hyperplane}\}.$$

$$\begin{array}{ccc} \mathbb{P}(V) & \xrightarrow{\sim} & |V| \\ \sim \downarrow & & \parallel \\ (\mathbb{P}(V^\vee))^\vee & \xrightarrow{H \mapsto \phi_V^*(H)} & |V|. \end{array}$$

Definition 5.2.4 (Separation and very ampleness). For $\mathcal{L} = \mathcal{O}_X(D)$,

$$\text{ev}_{p,q} : H^0(D) \longrightarrow \mathcal{L}|_p \oplus \mathcal{L}|_q, \quad s \longmapsto (s(p), s(q)) \quad (p \neq q).$$

For $z(p) = 0$ and a local frame of $\mathcal{L}$, the invariant first-jet map

$$j_p^1 : H^0(D) \longrightarrow H^0(2p, \mathcal{L}|_{2p}) \cong \mathbb{C}^2$$

is represented by

$$j_p^1(s) = \begin{bmatrix} s(p) \\ \frac{\partial}{\partial z} s(z) \Big|_{z=0} \end{bmatrix}.$$

Surjectivity of $\text{ev}_{p,q}$ separates points; surjectivity of j_p^1 separates tangent vectors. The divisor D is very ample when $\phi_{H^0(D)}$ is an embedding, and ample when mD is very ample for some $m \geq 1$.

Proposition 5.2.5 (Evaluation criteria). *For a length-two divisor*

$$Z := \begin{cases} p + q, & p \neq q, \\ 2p, & p = q, \end{cases}$$

the cohomology sequence is

$$0 \longrightarrow H^0(D - Z) \hookrightarrow H^0(D) \xrightarrow{\text{ev}_Z} H^0(Z, \mathcal{O}_X(D)|_Z)$$

$$\begin{array}{c} \swarrow \delta_Z \\ H^1(D - Z) \longrightarrow H^1(D) \longrightarrow 0. \end{array}$$

Consequently,

$$\ker(\text{ev}_p) = H^0(D - p), \quad \ker(\text{ev}_{p,q}) = H^0(D - p - q), \quad \ker(j_p^1) = H^0(D - 2p),$$

$$|D| \text{ is base-point free} \iff h^0(D - p) = h^0(D) - 1 \quad (p \in X),$$

$$D \text{ is very ample} \iff h^0(D - p - q) = h^0(D) - 2 \quad (p, q \in X),$$

where $p = q$ is tangent-vector separation.

Corollary 5.2.6 (High-degree systems).

$$\deg(D) \geq 2g \implies \deg(D - p) \geq 2g - 1 \xrightarrow{5.1.9} h^0(D - p) = h^0(D) - 1 \xrightarrow{5.2.5} |D| \text{ is base-point free},$$

$$\deg(D) \geq 2g + 1 \implies \deg(D - p - q) \geq 2g - 1 \xrightarrow{5.1.9} h^0(D - p - q) = h^0(D) - 2 \xrightarrow{5.2.5} D \text{ is very ample}.$$

Theorem 5.2.7 (Meromorphic functions and projective realization). *For every $p \in X$, there is a nonconstant meromorphic function holomorphic on $X \setminus \{p\}$. Moreover, X is a smooth projective algebraic curve and $X \setminus \{p\}$ is a smooth affine algebraic curve. Indeed,*

$$\begin{aligned} D = (2g+1)p &\xrightarrow{5.1.9} h^0(D) = g+2 > 1 \implies H^0(D) \setminus \mathbb{C} \neq \emptyset, \\ f \in H^0(D) \setminus \mathbb{C} &\implies \operatorname{div}(f) + D \geq 0 \implies f \in \mathcal{O}_X(X \setminus \{p\}), \quad \tilde{f}: X \rightarrow \mathbb{P}^1 \text{ nonconstant}, \\ \deg(D) = 2g+1 &\xrightarrow{5.2.6} D \text{ is very ample} \xrightarrow{5.2.4} \phi_D: X \hookrightarrow \mathbb{P}^{g+1}. \end{aligned}$$

For the hyperplane H corresponding to $1 \in H^0(D)$,

$$\begin{array}{ccc} X \setminus \{p\} & \xrightarrow{\quad} & X \\ \phi_D \downarrow & & \downarrow \phi_D \\ \phi_D(X) \setminus H & \xrightarrow{\quad} & \phi_D(X) \subset \mathbb{P}^{g+1}. \\ \phi_D^*(H) = D, & \mathbb{P}^{g+1} \setminus H \cong \mathbb{C}^{g+1}. \\ \phi_D(X) \subset \mathbb{P}^{g+1} \text{ is closed analytic} & \xrightarrow{A.0.2} & \phi_D(X) \text{ is projective algebraic.} \\ \phi_D(X) \setminus H \subset \mathbb{C}^{g+1} & \text{is affine algebraic.} \end{array}$$

Theorems 3.2.10 and 3.2.11 further give an embedding in $\mathbb{P}^3$ and a finite birational nodal plane model.

Definition 5.2.8 (Canonical map). For $g \geq 1$, choose a basis $\omega_1, \dots, \omega_g$ of $H^0(K_X) = H^0(X, \Omega_X)$. When $|K_X|$ is base-point free, define

$$(5.2.1) \quad \phi_{K_X}: X \longrightarrow \mathbb{P}^{g-1}, \quad p \longmapsto [\omega_1(p) : \dots : \omega_g(p)].$$

Proposition 5.2.9 (Canonical base-point freeness). *If $g \geq 2$, then $|K_X|$ is base-point free. Indeed,*

$$\begin{aligned} h^0(p) > 1 &\implies \exists f \in H^0(p) \setminus \mathbb{C} \implies \deg(\tilde{f}) = 1 \xleftrightarrow{2.2.6} X \cong \mathbb{P}^1, \\ g \geq 2 &\implies h^0(p) = 1, \quad h^0(K_X - p) - h^0(p) \stackrel{5.1.8}{=} \deg(K_X - p) - g + 1 = g - 2, \\ h^0(K_X - p) &= g - 1 = h^0(K_X) - 1 \xleftrightarrow{5.2.5} |K_X| \text{ is base-point free.} \end{aligned}$$

Theorem 5.2.10 (Canonical map dichotomy). *Let $g \geq 2$. If X is nonhyperelliptic, then ϕ_{K_X} embeds X as a smooth curve of degree $2g - 2$ in $\mathbb{P}^{g-1}$. If $\pi: X \rightarrow \mathbb{P}^1$ has degree 2, then*

$$\begin{array}{ccc} X & \xrightarrow{\phi_{K_X}} & \mathbb{P}^{g-1} \\ \pi \searrow & & \nearrow v_{g-1} \\ & \mathbb{P}^1 & \end{array} \quad v_{g-1}([u:v]) = [u^{g-1} : u^{g-2}v : \dots : v^{g-1}],$$

so ϕ_{K_X} is two-to-one onto the rational normal curve of degree $g - 1$.

$$\mathcal{O}_X(K_X) \cong \pi^* \mathcal{O}_{\mathbb{P}^1}(g-1).$$

Proof. Apply Riemann–Roch to $K_X - p - q$ and use the length-two evaluation criterion.

$$\begin{aligned} D = K_X - p - q &\stackrel{5.1.8, 5.1.9}{\implies} h^0(K_X - p - q) - h^0(p+q) = g - 3. \\ h^0(K_X - p - q) &= g - 2 \iff h^0(p+q) = 1 \xleftrightarrow{5.2.5} |K_X| \text{ separates } p+q. \\ h^0(p+q) \geq 2 &\implies \exists f \in H^0(p+q) \setminus \mathbb{C} : 1 < \deg(\tilde{f}) \leq 2 \xrightarrow{2.2.6} \deg(\tilde{f}) = 2 \implies X \text{ is hyperelliptic,} \\ X \text{ hyperelliptic} &\implies \exists p, q \in X : \pi^{-1}(\infty) = p+q, \quad h^0(p+q) = 2 \xrightarrow{5.2.5} |K_X| \text{ is not very ample.} \\ X \text{ nonhyperelliptic} &\stackrel{5.2.5}{\implies} \phi_{K_X} \text{ is an embedding,} \quad \deg(\phi_{K_X}(X)) = \deg(K_X) = 2g - 2. \end{aligned}$$

$$X = V(y^2 - h(x)), \quad \deg(h) \in \{2g+1, 2g+2\},$$

$$H^0(K_X) = \text{span}_{\mathbb{C}} \left(x^j \frac{dx}{y} : 0 \leq j \leq g-1 \right) \implies \phi_{K_X} = v_{g-1} \circ \pi.$$

Thus failure of length-two separation is precisely the hyperelliptic factorization. $\square$

Example 5.2.11 (Low-genus canonical models).

$$g = 0 : X \cong \mathbb{P}^1, \quad h^0(K_X) = 0; \quad g = 1 : K_X \sim 0, \quad \phi_{K_X} : X \rightarrow \mathbb{P}^0,$$

$$g = 2 : X \text{ is hyperelliptic}, \quad \deg(\phi_{K_X}) = 2,$$

$$g = 3, X \text{ nonhyperelliptic} : \phi_{K_X}(X) \subset \mathbb{P}^2 \text{ is a smooth plane quartic},$$

$$g = 4, X \text{ nonhyperelliptic} : \phi_{K_X}(X) = V(Q_2, Q_3) \subset \mathbb{P}^3, \quad \deg(Q_2) = 2, \quad \deg(Q_3) = 3.$$

For $g = 1$, every $3p$ satisfies

$$h^0(3p) = 3, \quad 3p \text{ is very ample}, \quad \phi_{3p}(X) \subset \mathbb{P}^2 \text{ is a smooth plane cubic.}$$

Proposition 5.2.12 (Multicanonical embeddings). *Let $g \geq 2$ and $n \geq 2$. Then*

$$\deg((1-n)K_X) < 0 \xrightarrow{5.1.5, 5.1.7} h^1(nK_X) = 0 \xrightarrow{5.1.8} h^0(nK_X) = (2n-1)(g-1).$$

Moreover,

$$\begin{aligned} g \geq 3 &\xrightarrow{2K_X \text{ very ample}} X \xrightarrow{\phi_{2K_X}} \mathbb{P}^{3g-4}, \\ g \geq 2 &\xrightarrow{3K_X \text{ very ample}} X \xrightarrow{\phi_{3K_X}} \mathbb{P}^{5g-6}. \end{aligned}$$

Indeed,

$$g \geq 3 \implies \deg(2K_X) = 4g - 4 \geq 2g + 1, \quad g \geq 2 \implies \deg(3K_X) = 6g - 6 \geq 2g + 1,$$

$$\deg(nK_X) \geq 2g + 1 \xrightarrow{5.2.6} nK_X \text{ is very ample.}$$

5.3. Abel–Jacobi Theory.

Theorem 5.3.1 (Normalized periods). *Choose the symplectic basis from Proposition A.0.1. Its intersection matrix is*

$$J = \begin{bmatrix} 0 & I_g \\ -I_g & 0 \end{bmatrix}, \quad \gamma \cdot \eta = \mathbf{m}_\gamma^\top \cdot J \cdot \mathbf{m}_\eta.$$

There is a unique basis $\omega_1, \dots, \omega_g$ of $H^0(X, \Omega_X)$ satisfying

$$A := \begin{bmatrix} \int_{a_1} \omega_1 & \cdots & \int_{a_1} \omega_g \\ \vdots & \ddots & \vdots \\ \int_{a_g} \omega_1 & \cdots & \int_{a_g} \omega_g \end{bmatrix} = I_g.$$

The period matrix

$$\tau := \begin{bmatrix} \int_{b_1} \omega_1 & \cdots & \int_{b_1} \omega_g \\ \vdots & \ddots & \vdots \\ \int_{b_g} \omega_1 & \cdots & \int_{b_g} \omega_g \end{bmatrix}$$

satisfies

$$\tau^\top = \tau, \quad \Im(\tau) = \frac{1}{2i}(\tau - \bar{\tau}) > 0, \quad 0 \neq \mathbf{v} \in \mathbb{C}^g \implies \bar{\mathbf{v}}^\top \cdot \Im(\tau) \cdot \mathbf{v} > 0.$$

Definition 5.3.2 (Jacobian). Define

$$\mathcal{P}: H_1(X, \mathbb{Z}) \longrightarrow H^0(X, \Omega_X)^\vee, \quad \mathcal{P}(\gamma)(\omega) := \int_\gamma \omega,$$

$$\Lambda_X := \text{im}(\mathcal{P}) = \mathbb{Z}^g + \tau \cdot \mathbb{Z}^g \subset \mathbb{C}^g.$$

Then

$$(5.3.1) \quad \text{Jac}(X) := H^0(X, \Omega_X)^\vee / \Lambda_X \cong \mathbb{C}^g / (\mathbb{Z}^g + \tau \cdot \mathbb{Z}^g).$$

The intrinsic and normalized-coordinate sequences are

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z}^{2g} & \xrightarrow{(\mathbf{m}, \mathbf{n}) \mapsto \mathbf{m} + \tau \cdot \mathbf{n}} & \mathbb{C}^g & \longrightarrow & \mathbb{C}^g / (\mathbb{Z}^g + \tau \cdot \mathbb{Z}^g) \longrightarrow 0 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & H_1(X, \mathbb{Z}) & \xrightarrow{\mathcal{P}} & H^0(X, \Omega_X)^\vee & \longrightarrow & \text{Jac}(X) \longrightarrow 0. \end{array}$$

Thus

$$\text{rank}_{\mathbb{Z}}(\Lambda_X) = 2g, \quad \dim_{\mathbb{C}}(\text{Jac}(X)) = g, \quad T_0 \text{Jac}(X) \cong H^0(X, \Omega_X)^\vee.$$

For $\lambda = \mathbf{m} + \tau \cdot \mathbf{n}$ and $\lambda' = \mathbf{m}' + \tau \cdot \mathbf{n}'$, set

$$E(\lambda, \lambda') := \mathbf{m}^\top \cdot \mathbf{n}' - \mathbf{n}^\top \cdot \mathbf{m}' \in \mathbb{Z}.$$

Theorem ?? identifies E as the unimodular Riemann form defining the principal polarization.

Definition 5.3.3 (Abel–Jacobi map). Fix $p_0 \in X$, a universal cover $\pi: \tilde{X} \rightarrow X$, and $\tilde{p}_0 \in \pi^{-1}(p_0)$. Define

$$\tilde{\mu}(\tilde{p}) := \begin{bmatrix} \int_{\tilde{p}_0}^{\tilde{p}} \pi^* \omega_1 \\ \vdots \\ \int_{\tilde{p}_0}^{\tilde{p}} \pi^* \omega_g \end{bmatrix} \in \mathbb{C}^g.$$

It descends through

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{\tilde{\mu}} & \mathbb{C}^g \\ \pi \downarrow & & \downarrow \\ X & \xrightarrow{\mu_{p_0}} & \text{Jac}(X). \end{array}$$

Extend linearly by

$$\mu_{p_0} \left(\sum_{j=1}^r n_j p_j \right) := \sum_{j=1}^r n_j \mu_{p_0}(p_j), \quad \text{Div}^0(X) := \{D \in \text{Div}(X) : \deg(D) = 0\}.$$

Proposition 5.3.4 (Well-definedness and differential). For $\gamma \in \pi_1(X, p_0)$, let T_γ be its deck transformation and $[\gamma]_{\text{ab}} \in H_1(X, \mathbb{Z})$ its homology class.

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{T_\gamma} & \tilde{X} \\ \tilde{\mu} \downarrow & & \downarrow \tilde{\mu} \\ \mathbb{C}^g & \xrightarrow{\mathbf{v} \mapsto \mathbf{v} + \mathcal{P}([\gamma]_{\text{ab}})} & \mathbb{C}^g. \end{array}$$

Hence

$$\tilde{\mu}(T_\gamma \tilde{p}) = \tilde{\mu}(\tilde{p}) + \mathcal{P}([\gamma]_{\text{ab}}) \implies \mu_{p_0}: X \longrightarrow \text{Jac}(X) \text{ is well-defined and holomorphic.}$$

The differential is the evaluation functional:

$$\begin{array}{ccccc} T_p X & \xrightarrow{d\mu_{p_0}|_p} & T_{\mu_{p_0}(p)} \text{Jac}(X) & \xrightarrow[\sim]{d(T_{-\mu_{p_0}(p)})} & T_0 \text{Jac}(X) \\ & \searrow v \mapsto (\omega \mapsto \omega_p(v)) & & & \downarrow \sim \\ & & & & H^0(X, \Omega_X)^\vee. \end{array}$$

Thus

$$\begin{aligned} & \left(d(T_{-\mu_{p_0}(p)}) \Big|_{\mu_{p_0}(p)} \cdot d\mu_{p_0}|_p(v) \right) (\omega) = \omega_p(v), \\ & \mathbb{P} \left(d(T_{-\mu_{p_0}(p)}) \Big|_{\mu_{p_0}(p)} \cdot d\mu_{p_0}|_p(T_p X) \right) = \phi_{K_X}(p). \end{aligned}$$

For $p_1 \in X$,

$$\mu_{p_1}(p) = \mu_{p_0}(p) - \mu_{p_0}(p_1), \quad \mu_{p_1}(D) - \mu_{p_0}(D) = -\deg(D) \cdot \mu_{p_0}(p_1),$$

so $\mu_{p_0}|_{\text{Div}^0(X)}$ is independent of p_0 .

Theorem 5.3.5 (Abel). *The Abel–Jacobi homomorphism satisfies*

$$\mu: \text{Div}^0(X) \rightarrow \text{Jac}(X), \quad D \in \text{Prin}(X) \iff \mu(D) = 0.$$

Equivalently,

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Prin}(X) & \hookrightarrow & \text{Div}^0(X) & \xrightarrow{\mu} & \text{Jac}(X) \longrightarrow 0 \\ & & & & \downarrow & & \parallel \\ & & & & \text{Pic}^0(X) & \xrightarrow{\sim} & \text{Jac}(X). \end{array}$$

Thus

$$(5.3.2) \quad \bar{\mu}: \text{Pic}^0(X) \xrightarrow{\sim} \text{Jac}(X), \quad [D] \mapsto \mu(D).$$

For $g \geq 1$, μ_{p_0} is an embedding, since

$$\mu_{p_0}(p) = \mu_{p_0}(q) \xLeftrightarrow{5.3.2} p - q \in \text{Prin}(X) \implies p = q,$$

where $p \neq q$ would produce a degree-1 map to $\mathbb{P}^1$ by Proposition 2.2.6; and

$$g \geq 2 \xRightarrow{5.2.9, 5.3.4} d\mu_{p_0}|_p \neq 0, \quad g = 1 \xRightarrow{5.1.3, 5.1.9} \text{div}(\omega) = 0 \implies d\mu_{p_0}|_p \neq 0.$$

$$\mu_{p_0} \text{ injective} \quad \wedge \quad d\mu_{p_0}|_p \neq 0 \xRightarrow{A.0.2} \mu_{p_0} \text{ is an embedding.}$$

Proposition 5.3.6 (Cohomological Picard variety). *The exponential sequence is*

$$0 \longrightarrow \mathbb{Z}_X \longrightarrow \mathcal{O}_X \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{O}_X^\times \longrightarrow 1.$$

Its cohomology segment is

$$0 \longrightarrow H^1(X, \mathbb{Z}) \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow \text{Pic}(X) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \longrightarrow 0.$$

Here

$$H^1(X, \mathcal{O}_X^\times) \cong \text{Pic}(X), \quad H^2(X, \mathbb{Z}) \cong \mathbb{Z}, \quad c_1(\mathcal{O}_X(D)) = \deg(D).$$

For $\alpha \in H^1(X, \mathbb{Z})$ and $\omega \in H^0(X, \Omega_X)$, normalize Poincaré and Serre duality by

$$\int_{\text{PD}(\alpha)} \omega = \int_X \omega \wedge \alpha^{0,1}.$$

Then Theorems 5.1.7 and 5.3.5 give

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^1(X, \mathbb{Z}) & \longrightarrow & H^1(X, \mathcal{O}_X) & \longrightarrow & \text{Pic}^0(X) \longrightarrow 0 \\ & & \text{PD} \downarrow \sim & & \sim \downarrow \text{SD} & & \sim \downarrow \bar{\mu} \\ 0 & \longrightarrow & H_1(X, \mathbb{Z}) & \xrightarrow{\mathcal{P}} & H^0(X, \Omega_X)^\vee & \longrightarrow & \text{Jac}(X) \longrightarrow 0. \end{array}$$

Therefore

$$\text{Pic}^0(X) \cong H^1(X, \mathcal{O}_X) / H^1(X, \mathbb{Z}) \cong H^0(X, \Omega_X)^\vee / \Lambda_X = \text{Jac}(X).$$

At the identity,

$$T_0 \text{Pic}^0(X) \xrightarrow{\sim} H^1(X, \mathcal{O}_X) \xrightarrow[5.1.7]{\sim} H^0(X, \Omega_X)^\vee \xrightarrow{\sim} T_0 \text{Jac}(X).$$

Definition 5.3.7 (Symmetric products and Brill–Noether loci). For $d, r \geq 0$, define

$$\begin{aligned} X^{(d)} &:= X^d / S_d, \quad a_d(p_1 + \cdots + p_d) := [\mathcal{O}_X(p_1 + \cdots + p_d)] \in \text{Pic}^d(X), \\ W_d^r(X) &:= \left\{ [\mathcal{L}] \in \text{Pic}^d(X) : h^0(X, \mathcal{L}) \geq r + 1 \right\}, \quad W_d(X) := W_d^0(X) = a_d(X^{(d)}). \end{aligned}$$

For every $[\mathcal{L}] \in \text{Pic}^d(X)$,

$$\begin{array}{ccc} \mathbb{P}(H^0(X, \mathcal{L})) & \xrightarrow{[s] \mapsto \text{div}(s)} & X^{(d)} \\ \downarrow & & \downarrow a_d \\ \{[\mathcal{L}]\} & \hookrightarrow & \text{Pic}^d(X). \end{array}$$

Thus

$$\begin{aligned} a_d^{-1}([\mathcal{L}]) &\cong \mathbb{P}(H^0(X, \mathcal{L})), \quad h^0(X, \mathcal{L}) > 0 \implies \dim_{\mathbb{C}}(a_d^{-1}([\mathcal{L}])) = h^0(X, \mathcal{L}) - 1, \\ h^0(X, \mathcal{L}) = 0 &\iff a_d^{-1}([\mathcal{L}]) = \emptyset. \end{aligned}$$

Theorem 5.3.8 (Jacobi inversion). Assume $g \geq 1$. Then

$$\mu^{(g)} : X^{(g)} \longrightarrow \text{Jac}(X), \quad p_1 + \cdots + p_g \longmapsto \mu(p_1) + \cdots + \mu(p_g),$$

and

$$\begin{array}{ccc} X^g & \longrightarrow & X^{(g)} \xrightarrow{a_g} \text{Pic}^g(X) \\ & & \mu^{(g)} \downarrow \quad \sim \downarrow T_{-gp_0} \\ & & \text{Jac}(X) \longleftarrow \sim \text{Pic}^0(X). \end{array}$$

Both maps issuing from $X^{(g)}$ are surjective and birational, where

$$T_{-gp_0}([\mathcal{L}]) := [\mathcal{L}(-gp_0)].$$

Proof. Identify the fibers with complete linear systems and apply Riemann–Roch in degree g .

$$\deg(\mathcal{L}) = g \xrightarrow{5.1.8} h^0(X, \mathcal{L}) = 1 + h^0(X, \mathcal{O}_X(K_X) \otimes_{\mathcal{O}_X} \mathcal{L}^\vee) \geq 1 \implies a_g \text{ is surjective.}$$

$$E \in a_g^{-1}([\mathcal{L}]) \xLeftrightarrow{5.1.6} E = \operatorname{div}(s), \quad [s] \in \mathbb{P}(H^0(X, \mathcal{L})) \implies a_g^{-1}([\mathcal{L}]) \cong \mathbb{P}(H^0(X, \mathcal{L})).$$

$$g = 1 \implies \deg(\mathcal{O}_X(K_X) \otimes_{\mathcal{O}_X} \mathcal{L}^\vee) = -1 \xrightarrow{5.1.5, 5.1.8} h^0(X, \mathcal{L}) = 1.$$

$$g \geq 2 \implies h^0(X, \mathcal{L}) > 1 \iff h^0(X, \mathcal{O}_X(K_X) \otimes_{\mathcal{O}_X} \mathcal{L}^\vee) > 0 \iff [\mathcal{L}] \in [K_X] - W_{g-2}(X).$$

$$g \geq 2, \quad \dim_{\mathbb{C}}(W_{g-2}(X)) \leq g-2 < g = \dim_{\mathbb{C}}(\operatorname{Pic}^g(X)) \xrightarrow{?, 5.3.1} h^0(X, \mathcal{L}) = 1 \quad \text{generically.}$$

Hence a_g , and therefore $\mu^{(g)}$, is surjective and birational. $\square$

Proposition 5.3.9 (Theta divisor). *For $g \geq 1$,*

$$\begin{array}{ccc} X^{(g-1)} & \xrightarrow{a_{g-1}} & W_{g-1}(X) \hookrightarrow \operatorname{Pic}^{g-1}(X) \\ & \searrow \bar{\mu} \circ T_{-(g-1)p_0} \downarrow & \sim \downarrow \bar{\mu} \circ T_{-(g-1)p_0} \\ & \Theta_{p_0} & \hookrightarrow \operatorname{Jac}(X). \end{array}$$

Here

$$\Theta_{p_0} := \bar{\mu}(T_{-(g-1)p_0}(W_{g-1}(X))) \subset \operatorname{Jac}(X), \quad \dim_{\mathbb{C}}(\Theta_{p_0}) = g-1,$$

and $\mathcal{O}_{\operatorname{Jac}(X)}(\Theta_{p_0})$ induces the principal polarization of Definition 5.3.2.

Example 5.3.10 (Extreme genera).

$$g = 0 \implies H^0(X, \Omega_X) = 0, \quad \operatorname{Jac}(X) = 0, \quad \operatorname{Pic}^0(X) = 0.$$

For $g = 1$,

$$\mu_{p_0}: X \xrightarrow{\sim} \operatorname{Jac}(X) \cong \mathbb{C}/(\mathbb{Z} + \tau \cdot \mathbb{Z}), \quad \Im(\tau) > 0.$$

Part 2. Curvature-Homogeneous Riemannian Geometry⁴

6. MOVING FRAMES AND THE CURVATURE REDUCTION

Let $V := \mathbb{R}^n$ have its standard inner product, let $G := O(V)$, and let $\pi: F \rightarrow M$ be the orthonormal frame bundle of a Riemannian manifold (M^n, g) . All tensor products, orthogonal complements, and adjoints in this part are taken over $\mathbb{R}$.

6.1. Curvature and the Bianchi Identities.

Definition 6.1.1 (Algebraic curvature tensors). For $r \in S^2(\Lambda^2 V^*)$, define

$$\beta(r)(x, y, z, w) := r(x, y, z, w) + r(y, z, x, w) + r(z, x, y, w).$$

The algebraic curvature representation is

$$\begin{aligned} \mathcal{K}_n := \mathcal{K}(V) &:= \ker(\beta), \quad \dim_{\mathbb{R}}(\mathcal{K}_n) = \frac{n^2(n^2-1)}{12}. \\ 0 &\longrightarrow \mathcal{K}_n \hookrightarrow S^2(\Lambda^2 V^*) \xrightarrow{\beta} \Lambda^4 V^* \longrightarrow 0. \end{aligned}$$

Equivalently,

$$r_{ijkl} = -r_{jikl} = -r_{ijlk} = r_{klij}, \quad r_{ijkl} + r_{iklj} + r_{iljk} = 0.$$

Definition 6.1.2 (Tautological and connection forms). The tautological form $\omega \in \Omega^1(F, V)$ and the Levi-Civita connection $\theta \in \Omega^1(F, \mathfrak{so}(V))$ are written

$$\omega = \begin{bmatrix} \omega^1 \\ \vdots \\ \omega^n \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta^1_1 & \cdots & \theta^1_n \\ \vdots & \ddots & \vdots \\ \theta^n_1 & \cdots & \theta^n_n \end{bmatrix}, \quad \theta^i_j + \theta^j_i = 0.$$

For matrix-valued forms, the dot includes matrix multiplication and exterior multiplication of the entries,

$$(\theta \cdot \omega)^i := \sum_{j=1}^n \theta^i_j \wedge \omega^j, \quad (\theta \cdot \theta)^i_j := \sum_{k=1}^n \theta^i_k \wedge \theta^k_j.$$

Proposition 6.1.3 (Cartan structure equations). Let $R: F \rightarrow \mathcal{K}_n$ be the curvature function. With $\Omega := d\theta + \theta \cdot \theta$,

$$(6.1.1) \quad d\omega = -\theta \cdot \omega, \quad d\theta = -\theta \cdot \theta + R(\omega \wedge \omega), \quad \Omega^i_j = \frac{1}{2} \sum_{k,l=1}^n R^i_{jkl} \omega^k \wedge \omega^l.$$

In components,

$$d \begin{bmatrix} \omega^1 \\ \vdots \\ \omega^n \end{bmatrix} = - \begin{bmatrix} \theta^1_1 & \cdots & \theta^1_n \\ \vdots & \ddots & \vdots \\ \theta^n_1 & \cdots & \theta^n_n \end{bmatrix} \cdot \begin{bmatrix} \omega^1 \\ \vdots \\ \omega^n \end{bmatrix}.$$

Proposition 6.1.4 (Bianchi identities). The structure equations give

$$\Omega^i_j \wedge \omega^j = 0 \iff R^i_{jkl} + R^i_{klj} + R^i_{ljk} = 0,$$

$$d\Omega + \theta \cdot \Omega - \Omega \cdot \theta = 0 \iff R^i_{jkl;m} + R^i_{jlm;k} + R^i_{jmk;l} = 0.$$

Equivalently, there is a function $R': F \rightarrow V^* \otimes \mathcal{K}_n$ satisfying

$$dR = -\theta \cdot R + \langle R', \omega \rangle, \quad \text{Alt}(R') = 0.$$

⁴Based on lectures by Robert L. Bryant [2].

$$d^2\omega = 0 \xrightarrow{(6.1.1)} \Omega \cdot \omega = 0$$

$$d\Omega + \theta \cdot \Omega - \Omega \cdot \theta = 0 \xrightarrow[\Omega=R(\omega \wedge \omega)]{} \text{Alt}(R') = 0$$

Proof. Differentiate Equation (6.1.1) and use the graded Leibniz rule.

$$0 = d^2\omega = -d\theta \cdot \omega + \theta \cdot d\omega = -(d\theta + \theta \cdot \theta) \cdot \omega = -\Omega \cdot \omega.$$

$$\Omega^i_j \wedge \omega^j = 0 \iff \sum_{j,k,l=1}^n R^i_{jkl} \omega^k \wedge \omega^l \wedge \omega^j = 0 \iff R^i_{jkl} + R^i_{klj} + R^i_{ljk} = 0.$$

$$d\Omega = d(d\theta + \theta \cdot \theta) = d\theta \cdot \theta - \theta \cdot d\theta = \Omega \cdot \theta - \theta \cdot \Omega.$$

$$d\Omega + \theta \cdot \Omega - \Omega \cdot \theta = 0 \iff \text{Alt}(R') = 0 \iff R^i_{jkl;m} + R^i_{jlm;k} + R^i_{jmk;l} = 0.$$

These are the algebraic and differential Bianchi identities. $\square$

Definition 6.1.5 (Curvature homogeneity). The metric g is curvature homogeneous of type $r \in \mathcal{K}_n$ when

$$\forall x, y \in M \quad \exists \iota_{xy}: T_x M \xrightarrow{\sim} T_y M, \quad \iota_{xy}^* g_y = g_x, \quad \iota_{xy}^* R_y = R_x.$$

Equivalently,

$$R(F) = G \cdot r, \quad B_r := R^{-1}(r) \neq \emptyset, \quad \pi(B_r) = M.$$

The implications among the three local conditions are

$$\nabla R = 0 \iff \text{locally symmetric} \implies \text{locally homogeneous} \implies \text{curvature homogeneous}.$$

Neither reverse implication holds in general for $n \geq 3$.

Proposition 6.1.6 (Orbit codimension). Let $H_r \subset G$ stabilize r and put $d_r := \dim_{\mathbb{R}}(H_r)$. Then

$$\dim_{\mathbb{R}}(G \cdot r) = \frac{n(n-1)}{2} - d_r,$$

$$\text{codim}_{\mathcal{K}_n}(G \cdot r) = \frac{n(n-1)(n-2)(n+3)}{12} + d_r.$$

Thus curvature homogeneity is an overdetermined, diffeomorphism-invariant second-order system for every $n > 3$.

6.2. The Cartan–Spencer System.

Proposition 6.2.1 (Curvature reduction). Use the action

$$(a \cdot r)(x_1, x_2, x_3, x_4) := r(a^{-1} \cdot x_1, a^{-1} \cdot x_2, a^{-1} \cdot x_3, a^{-1} \cdot x_4).$$

The curvature function is equivariant,

$$R(u \cdot a) = a^{-1} \cdot R(u), \quad H_r := \{a \in G : a \cdot r = r\}, \quad \mathfrak{h}_r := \{A \in \mathfrak{so}(V) : A \cdot r = 0\}.$$

$$\begin{array}{ccc} F \times G & \xrightarrow{(u,a) \mapsto u \cdot a} & F \\ R \times \text{id} \downarrow & & \downarrow R \\ \mathcal{K}_n \times G & \xrightarrow{(s,a) \mapsto a^{-1} \cdot s} & \mathcal{K}_n \end{array} \quad \begin{array}{ccc} B_r & \hookrightarrow & F \\ \downarrow & & \downarrow R \\ \{r\} & \hookrightarrow & \mathcal{K}_n. \end{array}$$

The level set B_r is the principal reduction

$$\begin{array}{ccccc}
H_r & \longrightarrow & B_r & \xrightarrow{\pi_r} & M \\
\downarrow & & \downarrow & & \parallel \\
G & \longrightarrow & F & \xrightarrow{\pi} & M.
\end{array}$$

Definition 6.2.2 (Bianchi tableau). For $A \in \mathfrak{so}(V)$, the infinitesimal action is

$$(A \cdot r)(x_1, x_2, x_3, x_4) := - \sum_{v=1}^4 r(x_1, \dots, A \cdot x_v, \dots, x_4).$$

Set

$$\begin{aligned}
\mathfrak{so}(V) &= \mathfrak{h}_r \oplus \mathfrak{m}_r, & \mathfrak{m}_r &:= \mathfrak{h}_r^\perp, & \theta|_{B_r} &= \psi + \tau, \\
\psi &:= \Pi_{\mathfrak{h}_r}(\theta|_{B_r}), & \tau &:= \Pi_{\mathfrak{m}_r}(\theta|_{B_r}).
\end{aligned}$$

The form ψ is an H_r -connection and τ is semibasic,

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathfrak{h}_r & \xrightarrow{\zeta_b} & T_b B_r & \xrightarrow{\omega_b} & V \longrightarrow 0 \\
& & \searrow 0 & & \downarrow \tau_b & & \swarrow p_b \\
& & & & \mathfrak{m}_r & &
\end{array}$$

Hence

$$\tau = p(\omega), \quad p: B_r \longrightarrow V^* \otimes \mathfrak{m}_r, \quad p(b \cdot h) = \text{Ad}(h^{-1}) \cdot p(b) \cdot h.$$

Define

$$\begin{aligned}
\beta_r: V^* \otimes \mathfrak{m}_r &\longrightarrow \Lambda^3 V^* \otimes \mathfrak{so}(V), \\
\beta_r(P)(x, y, z) &:= (P(x) \cdot r)(y, z) + (P(y) \cdot r)(z, x) + (P(z) \cdot r)(x, y), \\
A_r &:= \ker(\beta_r) \subset V^* \otimes \mathfrak{m}_r.
\end{aligned}$$

The second Bianchi identity yields

$$B_r \xrightarrow{p} A_r \hookrightarrow V^* \otimes \mathfrak{m}_r \xrightarrow{\beta_r} \Lambda^3 V^* \otimes \mathfrak{so}(V).$$

Proposition 6.2.3 (Reduced structure equations). For $A \in \mathfrak{h}_r$ and $P \in V^* \otimes \mathfrak{m}_r$, put

$$\begin{aligned}
(A \cdot P)(x) &:= [A, P(x)] - P(A \cdot x), & D^\Psi P &:= dP + \psi \cdot P, \\
D^\Psi \tau &:= d\tau + \psi \cdot \tau + \tau \cdot \psi, & \mathcal{C}_r(p) &:= r(\omega \wedge \omega) - p(\omega) \cdot p(\omega).
\end{aligned}$$

On B_r , Cartan's equations reduce to

$$\begin{aligned}
(6.2.1) \quad d\omega &= -\psi \cdot \omega - p(\omega) \cdot \omega, & d\psi &= -\psi \cdot \psi + \Pi_{\mathfrak{h}_r}(\mathcal{C}_r(p)), \\
D^\Psi(p(\omega)) &= \Pi_{\mathfrak{m}_r}(\mathcal{C}_r(p)).
\end{aligned}$$

The two curvature projections are encoded by

$$\begin{array}{ccc}
& d\theta + \theta \cdot \theta = r(\omega \wedge \omega) & \\
\Pi_{\mathfrak{h}_r} \swarrow & & \searrow \Pi_{\mathfrak{m}_r} \\
d\psi + \psi \cdot \psi = \Pi_{\mathfrak{h}_r}(\mathcal{C}_r(p)) & & D^\Psi(p(\omega)) = \Pi_{\mathfrak{m}_r}(\mathcal{C}_r(p)).
\end{array}$$

Proof. Project Equation (6.1.1) along $\mathfrak{so}(V) = \mathfrak{h}_r \oplus \mathfrak{m}_r$.

$$\begin{aligned} d\omega &= -(\psi + p(\omega)) \cdot \omega = -\psi \cdot \omega - p(\omega) \cdot \omega. \\ d\psi + d(p(\omega)) &= -\psi \cdot \psi - \psi \cdot p(\omega) - p(\omega) \cdot \psi - p(\omega) \cdot p(\omega) + r(\omega \wedge \omega). \\ \Pi_{\mathfrak{h}_r}(\psi \cdot p(\omega) + p(\omega) \cdot \psi) &= 0, \quad \Pi_{\mathfrak{m}_r}(\psi \cdot \psi) = 0. \\ d\psi + \psi \cdot \psi &= \Pi_{\mathfrak{h}_r}(r(\omega \wedge \omega) - p(\omega) \cdot p(\omega)). \\ d(p(\omega)) + \psi \cdot p(\omega) + p(\omega) \cdot \psi &= \Pi_{\mathfrak{m}_r}(r(\omega \wedge \omega) - p(\omega) \cdot p(\omega)). \end{aligned}$$

These are the reduced structure equations. $\square$

Definition 6.2.4 (Spencer obstruction). Since $p: B_r \rightarrow A_r$ is H_r -equivariant, there is $q: B_r \rightarrow V^* \otimes A_r$ such that

$$D^\psi p = q(\omega).$$

Define

$$\begin{aligned} \delta: V^* \otimes A_r &\longrightarrow \Lambda^2 V^* \otimes \mathfrak{m}_r, \quad \delta(q)(x, y) := q(x)(y) - q(y)(x), \\ \Phi_r(P)(x, y) &:= \Pi_{\mathfrak{m}_r}(r(x, y) - [P(x), P(y)]) + P(P(x) \cdot y - P(y) \cdot x). \end{aligned}$$

Then

$$(6.2.2) \quad dp = -\psi \cdot p + q(\omega), \quad \delta(q) = \Phi_r(p) = Q_r(p) + L_r.$$

The obstruction space and first prolongation are

$$\begin{array}{ccccccc} H^{0,2}(A_r) := \frac{\Lambda^2 V^* \otimes \mathfrak{m}_r}{\delta(V^* \otimes A_r)}, & A_r^{(1)} := \ker(\delta) \subset V^* \otimes A_r. & & & & & \\ 0 \longrightarrow A_r^{(1)} \hookrightarrow V^* \otimes A_r \xrightarrow{\delta} \Lambda^2 V^* \otimes \mathfrak{m}_r \longrightarrow H^{0,2}(A_r) \longrightarrow 0 & & & & & & \\ & & \uparrow \Phi_r & & & & \\ & & A_r & & & & \end{array}$$

The first algebraic compatibility condition is

$$\boxed{[\Phi_r(p)] = [Q_r(p) + L_r] = 0 \quad \text{in } H^{0,2}(A_r)},$$

not $Q_r(p) + L_r = 0$ in $\Lambda^2 V^* \otimes \mathfrak{m}_r$. Further compatibility is obtained by prolongation,

$$p \xrightarrow{D^\psi} q \xrightarrow{D^\psi} q_1 \xrightarrow{D^\psi} \dots \quad A_r \rightsquigarrow A_r^{(1)} \rightsquigarrow A_r^{(2)} \rightsquigarrow \dots$$

Corollary 6.2.5 (Vanishing tableau). *If $A_r = 0$, then every curvature-homogeneous realization of type r is locally symmetric,*

$$A_r = 0 \implies p = 0 \implies \tau = 0 \implies \nabla R = 0 \implies (M, g) \text{ is locally symmetric.}$$

More generally,

$$p = 0 \iff \nabla R = 0 \iff (M, g) \text{ is locally symmetric.}$$

7. LOW-DIMENSIONAL CURVATURE HOMOGENEITY

The curvature representation is rigid in dimension two, equivalent to the Ricci representation in dimension three, and splits into four irreducible pieces on oriented four-manifolds. These decompositions expose both the rigid and the genuinely inhomogeneous cases.

$$\alpha \circ \beta := \frac{1}{2}(\alpha \otimes \beta + \beta \otimes \alpha), \quad \alpha^2 := \alpha \circ \alpha \quad (\alpha, \beta \in \Lambda^k V^*).$$

7.1. Dimensions Two and Three.

Theorem 7.1.1 (Minding, [2]). *For a Riemannian surface (M^2, g) ,*

$$\mathcal{H}_2 \cong \mathbb{R}, \quad R_{ijkl} = K(g_{ik}g_{jl} - g_{il}g_{jk}), \quad H_r = \mathbf{O}(2), \quad A_r = 0,$$

curvature homogeneous $\iff K$ is constant $\iff \nabla R = 0 \iff$ locally a two-dimensional space form.

$$K > 0 \longrightarrow S_K^2 \quad K = 0 \longrightarrow \mathbb{R}^2 \quad K < 0 \longrightarrow \mathbb{H}_K^2.$$

Proposition 7.1.2 (Ricci determination in dimension three). *For $\dim_{\mathbb{R}}(V) = 3$, contraction is an isomorphism*

$$\begin{array}{ccccc} \mathcal{H}_3 & \xrightarrow[r \mapsto \text{Ric}(r)]{\sim} & S^2(V^*) & \xrightarrow{\sim} & \mathbb{R} \oplus S_0^2(V^*) \\ \in & & \in & & \in \\ r & \longmapsto & \text{Ric}(r) & \longmapsto & (\text{Scal}(r), \text{Ric}^\circ(r)). \end{array}$$

Its inverse is

$$(7.1.1) \quad R_{ijkl} = g_{ik}\text{Ric}_{jl} - g_{il}\text{Ric}_{jk} - g_{jk}\text{Ric}_{il} + g_{jl}\text{Ric}_{ik} - \frac{\text{Scal}(r)}{2}(g_{ik}g_{jl} - g_{il}g_{jk}).$$

Hence

$$(M^3, g) \text{ is curvature homogeneous} \iff \text{Spec}(\text{Ric}^\sharp) = \{c_1, c_2, c_3\} \text{ is constant on } M.$$

In an adapted orthonormal frame,

$$[\text{Ric}^\sharp] = \begin{bmatrix} c_1 & 0 & 0 \\ 0 & c_2 & 0 \\ 0 & 0 & c_3 \end{bmatrix}, \quad \text{Ric}^\sharp \cdot \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} c_1 e_1 \\ c_2 e_2 \\ c_3 e_3 \end{bmatrix}.$$

Corollary 7.1.3 (Three-dimensional stabilizers). *The multiplicity pattern of $\text{Ric}^\sharp$ determines H_r ,*

$$\begin{array}{c} c_1, c_2, c_3 \text{ distinct} \implies H_r \cong \mathbf{O}(1)^3 \\ \nearrow \\ \text{Spec}(\text{Ric}^\sharp) \longrightarrow c_1 = c_2 \neq c_3 \implies H_r \cong \mathbf{O}(2) \times \mathbf{O}(1) \\ \searrow \\ c_1 = c_2 = c_3 \implies H_r = \mathbf{O}(3). \end{array}$$

The final branch is the constant-sectional-curvature branch.

Theorem 7.1.4 (Kowalski–Prüfer, [2]). *For every constant triple (c_1, c_2, c_3) that is not scalar, there are local real-analytic, curvature-homogeneous, non-locally-homogeneous metrics with these Ricci eigenvalues. Modulo local diffeomorphism,*

$$c_1, c_2, c_3 \text{ distinct} \implies 3 \text{ functions of 2 variables,}$$

$$c_1 = c_2 \neq c_3 \implies 2 \text{ functions of 1 variable.}$$

In Ricci-diagonal geodesic normal coordinates, the corresponding k -jet counts are

$$\dim(\mathcal{J}_{c_1, c_2, c_3}^k) = 3 \binom{k}{2} = \frac{3}{2}k(k-1), \quad \dim(\mathcal{J}_{c, c, d}^k) = 2(k-1).$$

For distinct eigenvalues and an adapted coframe $\xi = \xi_1 \omega^1 + \xi_2 \omega^2 + \xi_3 \omega^3$, the characteristic variety modulo diffeomorphisms is

$$\Xi_r = \{[\xi] \in \mathbb{P}(V^*) : \xi_1 \xi_2 \xi_3 = 0\} = \{\xi_1 = 0\} \cup \{\xi_2 = 0\} \cup \{\xi_3 = 0\}.$$

$$\Sigma \subset M^3 \begin{array}{c} \xrightarrow{\omega^1 \wedge \omega^2 \neq 0} \\ \xrightarrow{\omega^1 \wedge \omega^3 \neq 0} \text{noncharacteristic Cauchy data} \Longrightarrow \text{local curvature-homogeneous metric} \\ \xrightarrow{\omega^2 \wedge \omega^3 \neq 0} \end{array}$$

For $c_1 = c_2 \neq c_3$, the transverse type is

$$\begin{array}{ccc} & & c_3 > 0 \Longrightarrow \text{elliptic} \\ & \nearrow & \\ c_1 = c_2 \neq c_3 & \longrightarrow & c_3 < 0 \Longrightarrow \text{hyperbolic} \\ & \searrow & \\ & & c_3 = 0 \Longrightarrow \text{degenerate.} \end{array}$$

Theorem 7.1.5 (Yamato, [2]). Let c_1, c_2, c_3 be distinct and satisfy

$$c_2 > 0 > c_1 + c_2 > c_3 \geq \frac{c_1^2 + c_2^2}{c_1 + c_2}.$$

There is a complete inhomogeneous metric on $\mathbb{R}^3$ with constant Ricci eigenvalues (c_1, c_2, c_3) , obtained from

$$g = (dz)^2 + E(z)(dx)^2 + 2F(z)dx \circ dy + G(z)(dy)^2, \\ [g]_{(dx, dy, dz)} = \begin{bmatrix} E(z) & F(z) & 0 \\ F(z) & G(z) & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad g = \begin{bmatrix} dx & dy & dz \end{bmatrix} \cdot [g]_{(dx, dy, dz)} \cdot \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}.$$

These metrics have cohomogeneity one under translation in x and y .

7.2. Dimension Four. Pass locally to oriented orthonormal frames. For a positively oriented coframe (e^1, e^2, e^3, e^4) , put

$$\eta_1^\pm := \frac{e^1 \wedge e^2 \pm e^3 \wedge e^4}{\sqrt{2}}, \quad \eta_2^\pm := \frac{e^1 \wedge e^3 \mp e^2 \wedge e^4}{\sqrt{2}}, \quad \eta_3^\pm := \frac{e^1 \wedge e^4 \pm e^2 \wedge e^3}{\sqrt{2}}.$$

$$[H]_{\Lambda_+^2 \oplus \Lambda_-^2} := \begin{bmatrix} I_3 & 0 \\ 0 & -I_3 \end{bmatrix}.$$

$$\begin{array}{ccc} \Lambda^2 V^* & \xrightarrow{\quad * \quad} & \Lambda^2 V^* \\ \sim \downarrow & & \downarrow \sim \\ \Lambda_+^2 V^* \oplus \Lambda_-^2 V^* & \xrightarrow{\alpha \mapsto H \cdot \alpha} & \Lambda_+^2 V^* \oplus \Lambda_-^2 V^*. \end{array}$$

Proposition 7.2.1 (Four-dimensional curvature operator). As an $\mathrm{SO}(4)$ -representation,

$$\mathcal{K}_4 \cong \mathbb{R} \oplus S_0^2(V^*) \oplus S_0^2(\Lambda_+^2 V^*) \oplus S_0^2(\Lambda_-^2 V^*), \quad 20 = 1 + 9 + 5 + 5.$$

$$\mathcal{K}_4 \xrightarrow{\sim} \mathbb{R} \oplus S_0^2(V^*) \oplus S_0^2(\Lambda_+^2 V^*) \oplus S_0^2(\Lambda_-^2 V^*)$$

$$\in$$

$$\in$$

$$r \longmapsto (\text{Scal}(r), \text{Ric}^\circ(r), W_r^+, W_r^-).$$

Relative to $(\eta_1^+, \eta_2^+, \eta_3^+, \eta_1^-, \eta_2^-, \eta_3^-)$, let $B = (b_{ij})$ encode $\text{Ric}^\circ(r)$. Then

$$(7.2.1) \quad [\mathcal{R}_r] = \begin{bmatrix} W_r^+ + \frac{\text{Scal}(r)}{12} \cdot I_3 & B \\ B^\top & W_r^- + \frac{\text{Scal}(r)}{12} \cdot I_3 \end{bmatrix}.$$

Explicitly, if $W_r^\pm = (w_{ij}^\pm)$,

$$[\mathcal{R}_r] = \begin{bmatrix} w_{11}^+ + \frac{s}{12} & w_{12}^+ & w_{13}^+ & b_{11} & b_{12} & b_{13} \\ w_{12}^+ & w_{22}^+ + \frac{s}{12} & w_{23}^+ & b_{21} & b_{22} & b_{23} \\ w_{13}^+ & w_{23}^+ & w_{33}^+ + \frac{s}{12} & b_{31} & b_{32} & b_{33} \\ b_{11} & b_{21} & b_{31} & w_{11}^- + \frac{s}{12} & w_{12}^- & w_{13}^- \\ b_{12} & b_{22} & b_{32} & w_{12}^- & w_{22}^- + \frac{s}{12} & w_{23}^- \\ b_{13} & b_{23} & b_{33} & w_{13}^- & w_{23}^- & w_{33}^- + \frac{s}{12} \end{bmatrix}, \quad s := \text{Scal}(r).$$

Orientation reversal interchanges W_r^+ and W_r^- .

Definition 7.2.2 (Einstein and conformally flat loci). The two linear loci are the kernels in

$$\begin{aligned} 0 &\longrightarrow \mathcal{K}_4^{\text{E}} \hookrightarrow \mathcal{K}_4 \xrightarrow{r \mapsto \text{Ric}^\circ(r)} S_0^2(V^*) \longrightarrow 0 \\ 0 &\longrightarrow \mathcal{K}_4^{\text{cf}} \hookrightarrow \mathcal{K}_4 \xrightarrow{r \mapsto (W_r^+, W_r^-)} S_0^2(\Lambda_+^2 V^*) \oplus S_0^2(\Lambda_-^2 V^*) \longrightarrow 0. \end{aligned}$$

Thus

$$r \in \mathcal{K}_4^{\text{E}} \iff B = 0, \quad r \in \mathcal{K}_4^{\text{cf}} \iff W_r^+ = W_r^- = 0.$$

Part 3. Appendix

APPENDIX A. FOUNDATIONAL TOOLS

Fact A.0.1 (Coverings and compact surfaces). *For a covering $\pi: (E, e_0) \rightarrow (B, b_0)$ and $f: (Z, z_0) \rightarrow (B, b_0)$,*

$$\tilde{f}: (Z, z_0) \rightarrow (E, e_0) \text{ exists} \iff f_*(\pi_1(Z, z_0)) \subset \pi_*(\pi_1(E, e_0)).$$

$$\begin{array}{ccc} [0, 1] \times \{0\} & \xrightarrow{\tilde{H}_0} & E \\ \downarrow & \searrow \tilde{H} & \downarrow \pi \\ [0, 1]^2 & \xrightarrow{H} & B \end{array} \quad \begin{array}{ccc} X^\circ & \hookrightarrow & X \\ F^\circ \downarrow & & \downarrow F \\ Y \setminus B & \hookrightarrow & Y. \end{array}$$

Every connected Riemann surface X has a universal cover and $X \cong \tilde{X}/\text{Deck}(\pi)$. If Y is compact, $B \subset Y$ is finite, and F° is finite, the second square has F unique over Y , with $F(z) = z^{e_p(F)}$ locally over B . For finite $S \subset \text{int}(M)$ and a d -sheeted cover $\tilde{M} \rightarrow M$,

$$\chi(M \setminus S) = \chi(M) - |S|, \quad \chi(\tilde{M}) = d \cdot \chi(M), \quad \pi_1(\mathbb{P}^1 \setminus \{q_1, \dots, q_r\}) \cong \langle \gamma_1, \dots, \gamma_r \mid \gamma_1 \cdots \gamma_r = 1 \rangle.$$

If X is compact of genus g , then

$$H_1(X, \mathbb{Z}) \cong \mathbb{Z}^{2g}, \quad J = \begin{bmatrix} 0 & I_g \\ -I_g & 0 \end{bmatrix}, \quad \gamma \cdot \eta = \mathbf{m}_\gamma^\top \cdot J \cdot \mathbf{m}_\eta, \\ g = 0 \iff X \cong \mathbb{P}^1, \quad g = 1 \iff X \cong \mathbb{C}/L.$$

Fact A.0.2 (Projective and intersection tools). *Projective algebraic sets are compact, irreducible projective sets are connected, and*

$$\mathbb{P}^n \setminus V(x_0) \xrightarrow{\sim} \mathbb{C}^n, \quad [x_0 : \dots : x_n] \mapsto \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right).$$

Every ideal in $\mathbb{C}[x_0, \dots, x_n]$ is finitely generated, and every closed analytic subset of $\mathbb{P}^n$ is projective algebraic. If $C = V(F)$ and $D = V(G)$ have no common component, then

$$I_p(C, D) := \dim_{\mathbb{C}} \left(\frac{\mathcal{O}_{\mathbb{P}^2, p}}{(f_p, g_p)} \right), \quad \deg(F) = m, \quad \deg(G) = n \implies \sum_{p \in C \cap D} I_p(C, D) = m \cdot n.$$

For a smooth curve $X \subset \mathbb{P}^N$,

$$\dim_{\mathbb{C}}(\text{Sec}(X)) \leq 3, \quad q \notin \text{Sec}(X) \implies \text{pr}_q|_X: X \hookrightarrow \mathbb{P}^{N-1}.$$

$$\begin{array}{ccc} X & \hookrightarrow & \mathbb{P}^N \setminus \{q\} \\ \text{pr}_q|_X \downarrow & & \downarrow \text{pr}_q \\ \text{pr}_q(X) & \hookrightarrow & \mathbb{P}^{N-1} \end{array} \quad X \text{ compact} \xrightarrow{F} Y \text{ Hausdorff} \implies F(X) \text{ closed}.$$

Thus every injective holomorphic immersion from a compact complex manifold is a closed embedding. If $X = V(P) \subset \mathbb{P}^2$ is smooth of degree d and $\pi_a([x_0 : x_1 : x_2]) = [x_0 : x_1]$ is generic, then

$$\deg(\pi_a) = d, \quad e_p(\pi_a) - 1 = I_p \left(X, V \left(\frac{\partial}{\partial x_2} P \right) \right), \quad \deg(R_{\pi_a}) = d \cdot (d - 1).$$

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